

Basic Principles of Medicine II

Neuroscience Practice Questions

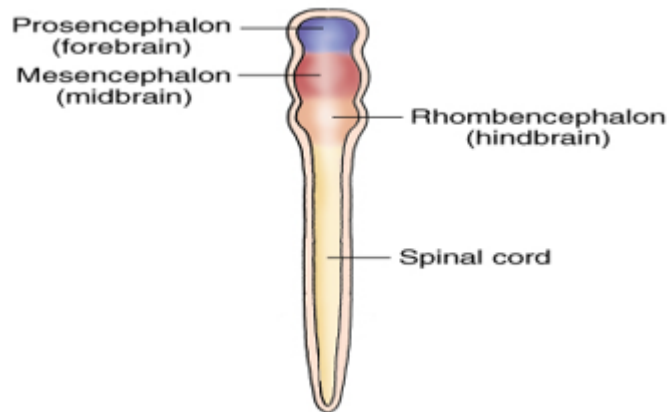
Lectures:

- **Neuroembryology**
- **General Morphology of the Nervous System**
- **Organisation of the Nervous System**
- **Meninges and Arteries**
- **Neurons and Glia**
- **Transport within the CNS**
- **Stimulated Excitable Cells**
- **Synaptic Transmission**

Questions Only

NB 01 – Neuroembryology

1. A 36-year-old woman in her fourth week of pregnancy comes to the physician to enquire about her baby's development. The development of the nervous system is explained to her. The development of which of the following is most likely induced by the notochord?
 - A. Neural tube
 - B. Neural fold
 - C. Neural plate
 - D. Neural crest
 - E. Neural sheath
2. The figure below shows the primary vesicles of the nervous system. From which of the following would the epithalamus most likely develop?



- A. Telencephalon
- B. Diencephalon
- C. Mesencephalon
- D. Myelencephalon
- E. Metencephalon

3. In the development of the forebrain, which of the following structures most likely develops from the telencephalon?

- A. Mammillary bodies
- B. Lateral ventricle
- C. Anterior pituitary
- D. Massa Intermedia
- E. Third ventricle

4. Cranial nerve nuclei situated more laterally in the brainstem develop from the _____ and are most likely _____ neurons.

- A. Floor plate, Sensory
- B. Basal plate, Sensory
- C. Basal plate, Motor
- D. Alar Plate, Sensory
- E. Alar Plate, Motor

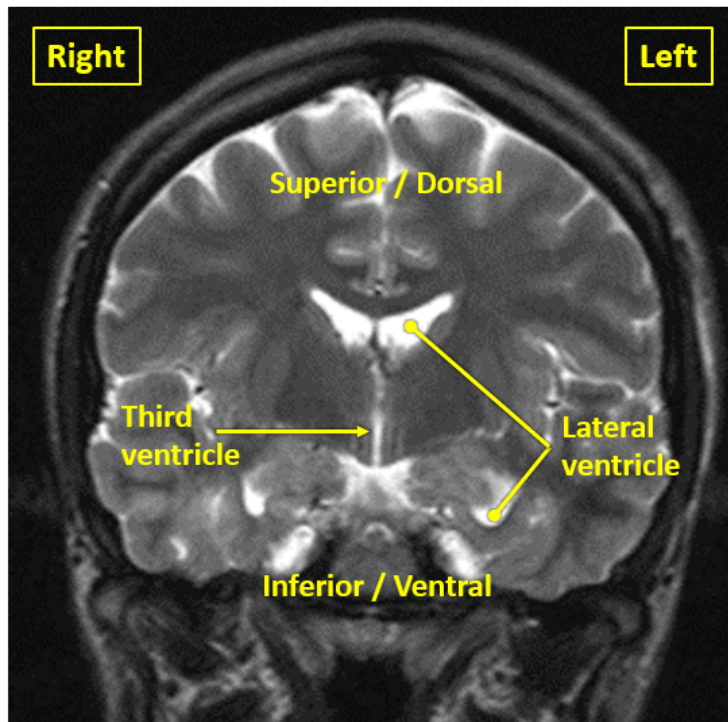
5. A newborn is being evaluated for congenital malformations of the brain. Imaging studies show a cystic enlargement of the fourth ventricle and an absence of the cerebellar vermis. Which of the following is the most likely condition in this patient?

- A. Anencephaly
 - B. Syringomyelia
 - C. Spina bifida
 - D. Dandy-Walker Syndrome
 - E. Encephalocele
6. A 23-year-old primigravida at 3 weeks of gestation learns that her embryo is undergoing neural tube development. As development progresses, the brainstem walls will reorganize so that sensory nuclei shift laterally while motor nuclei remain medial. This rearrangement most likely results from which developmental process?
- A. Formation of the cervical flexure
 - B. Formation of the pontine flexure
 - C. Migration of neural crest cells into the hindbrain
 - D. Rotation of the mesencephalon around the cephalic flexure
 - E. Expansion of the cerebral hemispheres and compression of the midbrain

NB 02 – General Morphology of the Nervous System

1. A 1-week-old child is brought to the physician because of a rapid increase in head size since birth. Examination and imaging studies are consistent with hydrocephalus due to aqueductal stenosis (narrowing of the cerebral aqueduct). The affected structure is most likely associated with which of the following?
- A. Diencephalon
 - B. Mesencephalon
 - C. Metencephalon
 - D. Myelencephalon
 - E. Telencephalon
2. A 50-year-old woman comes to the physician because of a headache that has lasted for 3 days. Vital signs are within normal limits. Examination findings show no abnormality. Imaging studies are

subsequently done for the patient. Which of the following is the most likely orientation of the image below?



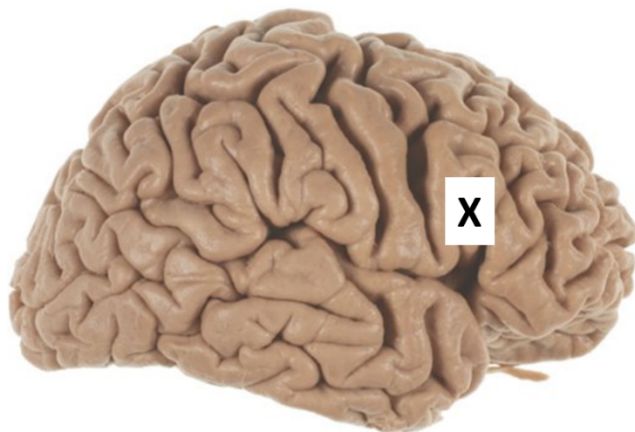
- A. Sagittal
 - B. Transverse
 - C. Horizontal
 - D. Coronal
 - E. Axial
3. A 72-year-old man is brought to the physician because of difficulty in understanding spoken language that has lasted for 2 days. Examination findings and imaging are consistent with sensory aphasia secondary to a stroke. The affected lobe lies inferior to the lateral sulcus. Which of the following lobes is most likely affected in this patient?
- A. Parietal
 - B. Frontal
 - C. Occipital
 - D. Insular

E. Temporal

4. A 57-year-old man comes to the physician because of a 2-week history of tingling sensations on the left side of his body. Neurological examination shows general somatosensory loss affecting the left side of the face and the left hand. Which of the following lobes is most likely lesioned?
- A. Left frontal
 - B. Left temporal
 - C. Left occipital
 - D. Right parietal
 - E. Right occipital
 - F. Right frontal
 - G. Left parietal
 - H. Right temporal
5. A 41-year-old woman comes to the physician because of a 12-hour history of clumsiness. Strength appears to be normal for all limbs. However, she lacks coordination with the right arm and leg. Imaging studies show a lesion in a structure that arises developmentally from the dorsal aspect of the pons. Which of the following structures is most likely injured?
- A. Spinal cord
 - B. Medulla
 - C. Midbrain
 - D. Thalamus
 - E. Cerebellum
6. A 44-year-old man is brought to the emergency department because of a 6-day history of worsening headaches, nausea, and vomiting. Imaging shows dilation of the third ventricle and both lateral ventricles. Which of the following ventricular passages is most likely blocked?

- A. Right interventricular foramen
- B. Left interventricular foramen
- C. Interventricular foramina
- D. Cerebral aqueduct
- E. Foramen of Magendie
- F. Foramina of Luschka

7. A 6-year-old boy is brought to the physician by his mother because of a 2-month history of headaches and difficulty concentrating. The headaches are localized to the right side of the head. MRI of the brain shows a tumor residing at the location indicated by the 'X' in the image of a brain specimen below. If the tumor growth progresses ventrally, which of the following structures will most likely be affected?



[Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 11e](#) Figure 2-11, modified

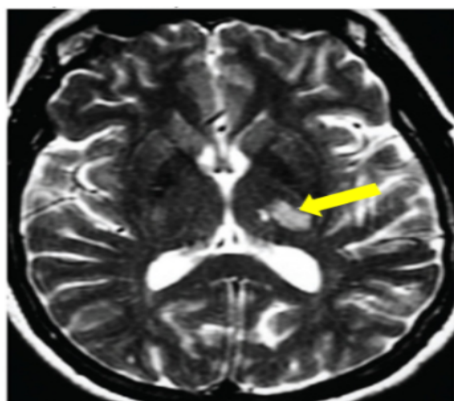
- A. Supramarginal gyrus
 - B. Middle frontal gyrus
 - C. Postcentral gyrus
 - D. Precentral gyrus
 - E. Superior temporal gyrus
8. A 51-year-old woman comes to the physician because of a 2-day history of difficulty looking upwards voluntarily. Imaging studies show a lesion affecting the superior colliculi. Which of the following brain regions are the affected structures most likely located?

- A. Ventral midbrain
- B. Dorsal midbrain
- C. Ventral pons
- D. Dorsal pons
- E. Ventral medulla

9. A 39-year-old man is brought to the emergency department because of a 3-hour history of loss of sensation on the right side of his body. Neurologic examination shows loss of touch and vibration sensations on the right side of his body. Imaging studies show a left diencephalic lesion. Which of the following diencephalic structures is most likely affected in this patient?

- A. Epithalamus
- B. Hypothalamus
- C. Subthalamus
- D. Thalamus

10. A 56-year-old woman is brought to the emergency department by her son because of a 4-hour history of numbness. The woman is drowsy on arrival but is oriented to time, place, and person. She suffered sudden onset of headache followed by numbness in her left hand. She has a 30-year history of hypertension and does not regularly take her medication. She admits regular, heavy consumption of alcohol. T2-weighted MRI of the brain shows a unilateral hyperintensity indicative of an intracerebral haemorrhage in the posterior diencephalon, indicated by the yellow arrow in the attached image. With which of the following components of the ventricular system is the affected structure most likely associated?



- A. Left lateral ventricle
- B. Right lateral ventricle
- C. Cerebral aqueduct
- D. Third ventricle
- E. Fourth ventricle

11. A 17-year-old boy is brought to the emergency department having hit his head while colliding with another player during a football game. He had a brief loss of consciousness immediately following the incident. He is alert and oriented on arrival. Neurological examination is normal. CT scan of the head does not show any evidence of intracranial bleeding. However, an incidental finding of a neoplasm deriving from a midline diencephalic structure that forms part of the posterior wall of the third ventricle is made. In which of the following structures is the neoplasm most likely located?

- A. Thalamus
- B. Pituitary gland
- C. Pineal gland
- D. Mammillary bodies
- E. Hypothalamus

NB 03 – Organisation of the Nervous System

1. A 64-year-old man comes to the physician because of a 6-hour history of weakness affecting the left hand. Neuroimaging studies show a small cortical lesion within the frontal lobe. Which of the following structures most closely overlaps with the site of the injury?

- A. Motor nucleus
- B. Motor tract
- C. Somatosensory nucleus
- D. Somatosensory tract

2. A 25-year-old woman is brought to the physician because of an inability to pronounce words correctly for 3 weeks. On examination, she is unable

to repeat words or sentences. Imaging studies show damage to an association fiber, and she is subsequently diagnosed with conduction aphasia. Which of the following fibers is most likely damaged in this patient?

- A. Internal capsule
- B. Corona radiata
- C. Corpus callosum
- D. Arcuate fasciculus
- E. Anterior commissure

3. A 20-year-old man is brought to the emergency room following a car crash about 20 minutes ago. He is unconscious. Imaging shows an epidural hematoma, with herniation of the uncus and compression of the midbrain. Which of the following cranial nerves is most likely affected?

- A. Olfactory
- B. Optic
- C. Oculomotor
- D. Facial
- E. Vestibulocochlear

4. A 32-year-old man is brought to the emergency room by the paramedics following a motor vehicle accident. Examination and investigation findings were consistent with a right hemisection (damage to the right side) of the spinal cord at the C5 vertebral level. Which of the following neurological deficits would this patient most likely have?

- A. Paralysis of both upper limbs
- B. Paralysis of both lower limbs
- C. Paralysis of the left upper limb
- D. Paralysis of the left lower limb
- E. Paralysis of the right lower limb

5. A 50-year-old woman comes to the physician because of a headache and double vision that have lasted for 3 days. On neurologic examination, she is unable to move her right eye laterally. Which of the following cranial nerves is most likely affected in this patient?

- A. Olfactory
- B. Optic
- C. Abducens
- D. Facial
- E. Vestibulocochlear

6. A 53-year-old man comes to the physician because of a 3-month history of memory problems. Imaging studies show a unilateral hippocampal lesion. Which of the following lobes of the cerebrum is most likely affected in this patient?

- A. Frontal
- B. Parietal
- C. Insular
- D. Occipital
- E. Temporal

7. A 64-year-old woman comes to the physician because of a 1-month history of double vision. Imaging studies show a dorsal brainstem lesion. Which of the following cranial nerves is most likely affected in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V
- F. VI
- G. VII
- H. VIII

- I. IX
- J. X
- K. XI
- L. XII

8. A 64-year-old man comes to the physician because of a 2-week history of headaches. MRI of the head shows a tumor located laterally at the right ponto-medullary junction. Which of the following cranial nerves is most likely affected in this patient?

- A. Olfactory
- B. Optic
- C. Oculomotor
- D. Trochlear
- E. Trigeminal
- F. Abducens
- G. Facial
- H. Vestibulocochlear
- I. Glossopharyngeal
- J. Vagus
- K. Accessory
- L. Hypoglossal

NB 04 – Meninges and Arteries

1. A 13-year-old girl is brought to the emergency department by her parents because of a 12-hour history of rising temperature, photophobia, a stiff neck, and irritability. Laboratory studies involve the drawing of CSF. Between which of the following tissue layers is the sample most likely drawn?

- A. Skull and periosteal dura
- B. Periosteal dura and meningeal dura

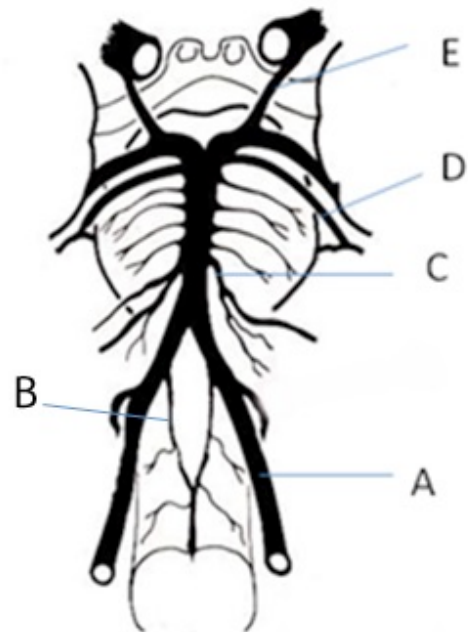
- C. Meningeal dura and arachnoid
- D. Arachnoid and pia
- E. Pia and spinal cord

2. A 36-year-old man is brought to the emergency department by paramedics because of a 1-hour history of trauma to the left side of the head. He lost consciousness at the moment of impact but regained consciousness within 5 minutes. The skin on the left side of his head is split. Six hours later, he becomes lethargic and shows signs of cerebral compression from an accumulating mass of blood. Which of the following arteries is most likely ruptured?

- A. Middle cerebral
- B. Posterior cerebral
- C. Middle meningeal
- D. Anterior cerebral
- E. Internal carotid
- F. Vertebral
- G. Basilar

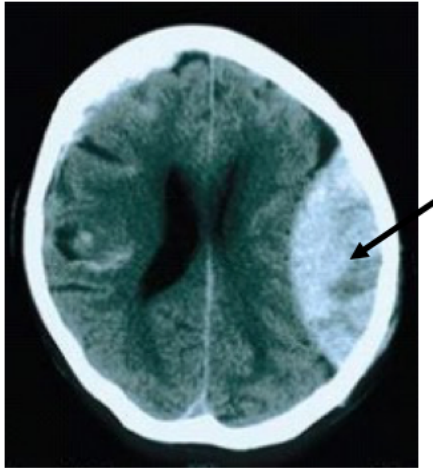
3. A 55-year-old man is brought to the emergency department because of a 2-hour history of loss of balance, nausea, vomiting, and a sensation that the room is spinning. A lesion situated at the junction of the medulla and pons is suspected. Which of the following subarachnoid arteries is most likely implicated in the injury (see attached image)?

- A. A
- B. B
- C. C
- D. D
- E. E



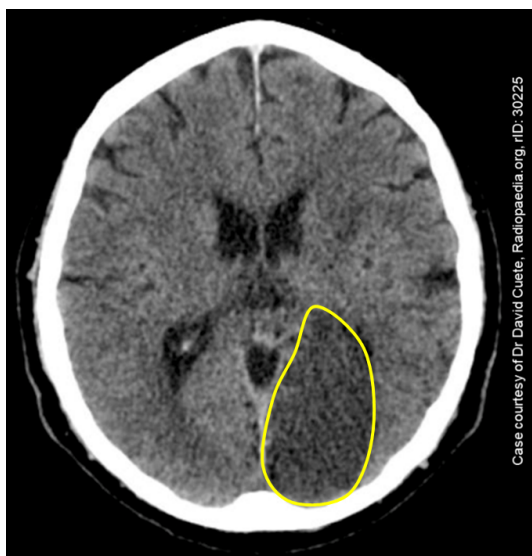
4. A 69-year-old woman comes to the physician because of a 2-week history of headache and confusion following minor head trauma. Imaging studies of the head show a right-sided crescent-shaped hemorrhage. Rupture of which of the following vessels is/are the most likely damaged in this patient?
 - A. Anterior communicating artery
 - B. Anterior cerebral artery
 - C. Bridging veins
 - D. Lenticulostriate artery
 - E. Middle meningeal artery

5. A 24-year-old man is brought to the emergency department following a motor vehicle accident 30 minutes ago. Imaging studies show an intracranial hemorrhage. Which of the following is the most likely hemorrhage depicted by the arrow (see attached image)?



- A. Epidural
- B. Subdural
- C. Subarachnoid
- D. Intracerebral

6. A 57-year-old woman is brought to the emergency department because of visual problems that began 8 hours previously. Neurologic examination shows right visual field deficits. CT scan of the head shows an infarction (outlined in yellow in the image). Identify the cerebral artery most likely occluded and the gyrus affected.



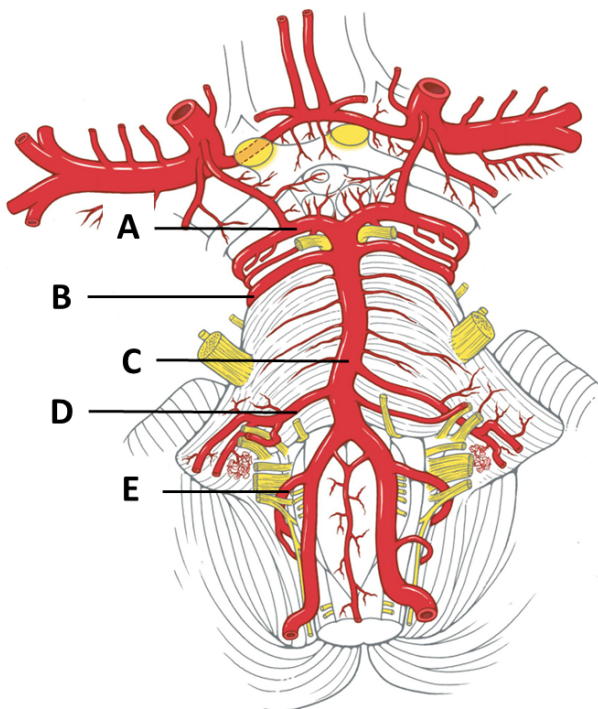
- A. Left anterior, posterior paracentral
- B. Left middle, postcentral

- C. Left posterior, lingual
- D. Right anterior, anterior paracentral
- E. Right middle precentral
- F. Right posterior, cuneus

7. A 52-year-old man is brought to the emergency department because of dizziness and nausea that began 6 hours ago. Neurologic examination shows somatosensory and cranial nerve deficits that are consistent with a dorsolateral medullary lesion. Which of the following arteries is most likely occluded?

- A. Posterior cerebral
- B. Basilar
- C. Anterior spinal
- D. Posterior inferior cerebellar
- E. Superior cerebellar

8. A 68-year-old man is brought to the emergency department because of weakness that began 12 hours ago. Neurologic examination shows motor, somatosensory, and cranial nerve deficits that are consistent with a ventral midbrain lesion. Which of the following arteries is most likely occluded?



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 11e Figure 2-21, modified

9. A 62-year-old man is brought to the emergency department because of a 1-hour history of a sudden-onset of severe headache and nuchal rigidity. CT scan of the head shows diffuse blood within the sulci and basal cisterns, outlining the convolutions of the brain. Which of the following vessels is/are most likely ruptured in this patient?
- A. Middle meningeal artery
 - B. Lenticulostriate artery
 - C. Bridging veins
 - D. External carotid artery
 - E. Posterior communicating artery
10. A 28-year-old man comes to the physician because of a 3-month history of nausea and headaches. MRA of the head shows an arteriovenous malformation involving major subarachnoid vessels supplying the insular cortex and lateral cerebral surface. Which of the following arteries is most likely implicated in this patient?
- A. Anterior cerebral
 - B. Posterior cerebral
 - C. Middle cerebral
 - D. Basilar
 - E. Vertebral
11. A 58-year-old man is brought to the emergency department because of a 4-hour history of confusion and a loss of awareness of the left side of space. CT scan of the head shows a hemorrhage arising from a malformation of the middle cerebral artery that follows the cortical sulci of the right lateral parietal cortical surface. The partially clotted mass is not restricted to the anatomical boundaries of the skull and assumes neither a lenticular nor crescent-like shape. Which of the following compartments is most likely filling with blood in this patient?
- A. Epidural space
 - B. Subdural space

- C. Subarachnoid space
- D. Periventricular space
- E. Ventricular system

12. A 50-year-old man is brought to the emergency department because of a 1-hour history of a sudden headache. CT scan of the head confirms a subarachnoid hemorrhage. Which of the following layers prevents the blood from contacting the surface of the brain in this patient?

- A. Dura mater
- B. Arachnoid
- C. Pia mater
- D. Periosteal dura
- E. Ligamentum denticulatum

13. A 30-year-old woman comes to the physician because of a 3-hour history of visual problems. Neurologic examination shows a palsy related to dysfunction of the oculomotor nerve. Subsequent imaging reveals an aneurysm originating from the circle of Willis, specifically at the junction of the posterior communicating artery. Which of the following anatomical spaces would most likely fill with blood in the event of a rupture of the aneurysm in this patient?

- A. Epidural space
- B. Subdural space
- C. Subarachnoid space
- D. Intraparenchymal space
- E. Cavernous sinus

14. During a craniotomy, a neurosurgeon encounters the outermost, tough, fibrous layer of the cranial meninges. This layer forms two principal reflections or folds that partially separate parts of the brain. Which of the following structures is/are most likely under observation in this surgical patient?

- A. Denticulate ligaments
- B. Filum terminale

- C. Falx cerebri
- D. Arachnoid villi
- E. Choroid plexus

15. A 42-year-old woman comes to the physician because of a 2-hour history of trauma to the side of her head. She does not know whether she was unconscious after the impact. CT of the head shows a fracture of the temporal bone and an associated bleed. Which of the following structures is most likely related to the damaged vessel in this patient?

- A. Pia mater
- B. Arachnoid
- C. Dura mater
- D. Leptomeninges
- E. Choroid plexus

16. A 72-year-old patient is brought to the physician because of a 3-month history of cognitive decline. He has a 5-year history of recurrent meningitis. MRI of the head shows enlarged ventricles, but no obstruction within the ventricular system or its major outflow foramina. Which of the following structures is most likely dysfunctional in this patient?

- A. Pia mater
- B. Choroid plexus
- C. Arachnoid villi
- D. Denticulate ligaments
- E. Ligamentum flavum
- F. Cerebral aqueduct

17. A 20-year-old man is brought to the emergency department because of a 1-hour history of a traumatic injury involving deceleration of the head. CT scan of the head shows brain damage associated with blood leaking from the anterior cerebral artery. Which of the following structures are most likely damaged in this patient?

- A. Brainstem and cerebellum

- B. Temporal lobe and Sylvian fissure
- C. Frontal and parietal lobes
- D. Occipital lobe and visual cortex
- E. Basal ganglia and thalamus

18. During a neurosurgical procedure, a surgeon notes a small growth overlapping with a denticulate ligament. Which of the following descriptions best applies to the cellular and anatomical features of the affected structure in this patient?

- A. Nervous; lateral to spinal cord
- B. Meningeal; lateral to spinal cord
- C. Nervous; caudal to spinal cord
- D. Meningeal; caudal to spinal cord
- E. Nervous; rostral to spinal cord
- F. Meningeal; rostral to spinal cord

19. A 54-year-old man is brought to the emergency department because of a 20-minute history of unconsciousness. Physical examination shows ataxic breathing, consistent with damage to the caudal brainstem. CT scan of the head shows a subarachnoid hemorrhage. Which of the following arteries has most likely ruptured in this patient?

- A. Anterior cerebral
- B. Middle cerebral
- C. Internal carotid
- D. External carotid
- E. Posterior spinal

20. A neurosurgeon is planning to access a tumor located within the brain parenchyma. After incising the outer meninges, the surgeon encounters numerous small blood vessels closely adherent to the surface of the brain. Which of the following structures most likely covers the vessels under observation in this patient?

- A. Periosteal dura
- B. Meningeal dura

- C. Pia mater
- D. Arachnoid
- E. Falx cerebri

21. A 31-year-old woman is brought to the emergency department because of a 30-minute history of traumatic injuries stemming from a motor vehicle collision. A whiplash injury is diagnosed with involvement of the blood supply to the ventral myelon. Which of the following arteries is most likely affected in this patient?

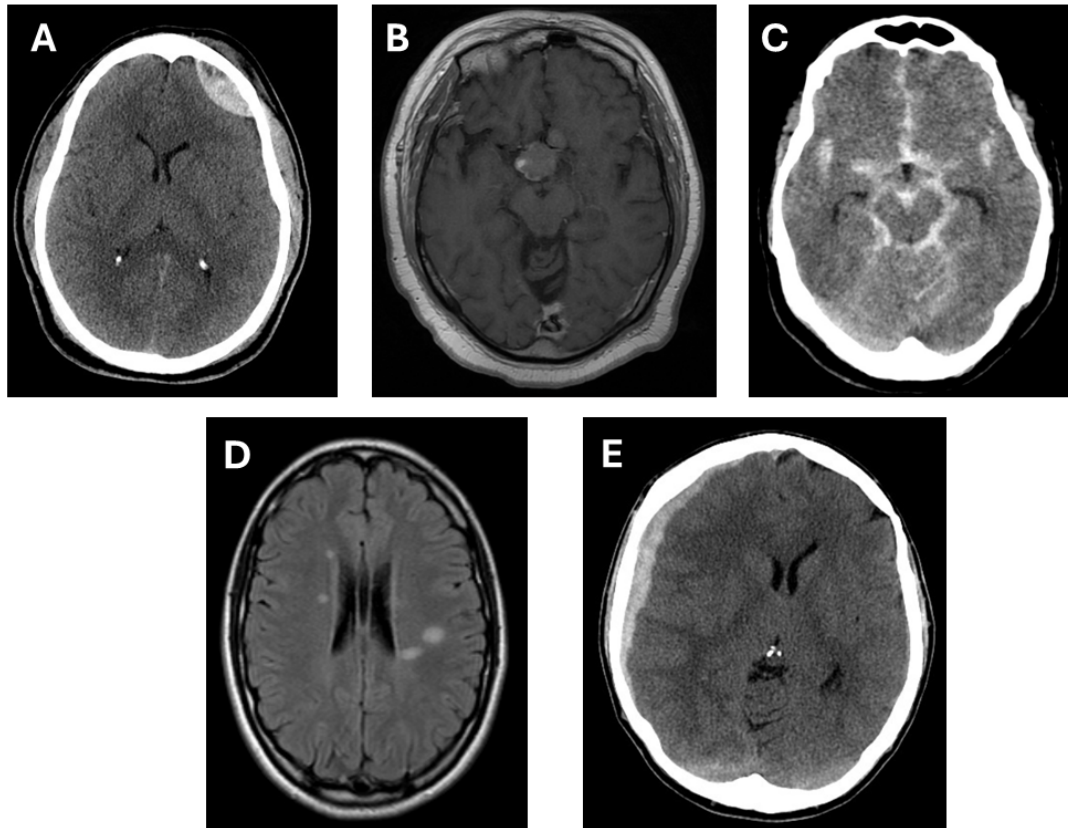
- A. Vertebral
- B. Posterior spinal
- C. Anterior spinal
- D. Posterior inferior cerebellar
- E. Anterior inferior cerebellar

22. A 43-year-old man is brought to the emergency department by ambulance 2 hours after his husband found him lying at the foot of the stairs in a confused state. On arrival at the hospital, the patient is unconscious. Shortly after arrival, the patient briefly regains consciousness but loses consciousness again before the neurological examination can be completed. A swelling is evident in the right temporal region. CT scan of the head shows a right temporoparietal hyperdense, biconvex extra-axial lesion that does not cross the suture lines of the skull. Which of the following findings would most likely have been observed if neurological examination had been completed?

- A. Weakness on the right side of the body
- B. Inability to comprehend spoken language
- C. Non-fluent speech
- D. Poor impulse control
- E. Reduced somatosensation on the left

23. A 72-year-old man is brought to the physician by his wife because of a 5-week history of forgetfulness. His wife reports that her husband was

previously meticulously organized and never forgot arrangements or belongings, but has recently had trouble remembering and finding things. The man recalls falling from a standing position approximately 6 weeks ago. He does not think he hit his head and did not require any medical attention. Neuroimaging shows a right-sided subdural hemorrhage. Which of the following images is most likely taken from this patient?



Images from <https://radiopaedia.org/>; full source details provided in the associated 'With Answers' document

NB 05 – Neurons and Glia

1. A 23-year-old man is brought to the emergency room because of a 4-hour history of unconsciousness. He dies 20 minutes later. Post-mortem studies reveal derangement of cells that normally contribute to the blood-brain barrier. Which of the following glial cells are most likely under study?
 - A. Astrocytes
 - B. Bipolar
 - C. Ependymal

- D. Microglia
- E. Oligodendrocytes
- F. Schwann

2. A neuroscience professor investigates the relationship between axonal transport and the onset of neurodegeneration. A defect in the anterograde axonal transport of which of the following is most likely implicated?
- A. Cytoskeletal molecules
 - B. Lysosomes
 - C. Recycled vesicular membranes
 - D. Nerve growth factor
3. A 54-year-old woman comes to the physician because of numbness in both feet that has lasted for 2 days. She has a history of Type 2 diabetes mellitus. Neurological examination shows loss of touch and vibration sensation on both feet. Cells of which of the following morphotypes most likely affected in this patient?
- A. Pseudo-unipolar
 - B. Unipolar
 - C. Bipolar
 - D. Multipolar
4. In an experimental setting, axonal transport is assessed. Which of the following structures is most likely responsible for retrograde axonal transport?
- A. Actin
 - B. Dynein
 - C. Kinesin
 - D. Neurofilament
 - E. Ribosome

5. A 68-year-old man comes to the physician because of a 4-week history of rapidly progressing weakness and muscle atrophy, initially in his hands. Neurologic examination shows weak myotatic reflexes. Clinical studies show a disease process primarily attacking spinal gray matter. Which of the following neuronal structures is most likely degenerating in this patient?

- A. Dendrite
- B. Soma
- C. Axon
- D. Axon terminal
- E. Nucleus

6. A 4-year-old boy is brought to the physician because of a 1-year history of developmental delays, seizures, and facial abnormalities. Genetic testing shows a disorder that results in the accumulation of undegraded macromolecules. Which of the following organelles is most likely dysfunctional in this patient?

- A. Mitochondrion
- B. Endoplasmic Reticulum
- C. Golgi Apparatus
- D. Lysosome
- E. Peroxisome

7. A 72-year-old woman is brought to the physician because of a 6-month history of declining memory. She receives a diagnosis of Alzheimer's disease, which features impaired axonal transport, leading to dysfunctional neurotransmitter delivery to synaptic terminals. Which of the following structures is/are most likely deranged in this patient?

- A. Neurofilaments
- B. Microtubules
- C. Microfilaments
- D. Rough endoplasmic reticulum
- E. Cytoplasm

8. A neuroscientist is studying a novel cultured cortical interneuron that receives numerous inputs from various surrounding neurons and integrates these signals. This neuron exhibits a highly branched dendritic tree and a relatively short axon. What is the most likely morphological classification of this neuron?
- A. Bipolar
 - B. Pseudounipolar
 - C. Multipolar
 - D. Pyramidal
 - E. Unipolar
9. A 28-year-old woman comes to the physician because of a 3-year history of vision loss, numbness, and weakness. Her neurological symptoms are attributed to demyelination in the central nervous system. Which of the following processes is most likely impaired in this patient?
- A. Propagation of action potentials
 - B. Synaptic transmission
 - C. Anterograde axonal transport
 - D. Diffusion
 - E. Retrograde axonal transport
10. A 33-year-old woman comes to the physician because of a 2-hour history of a deep penetrating wound to her left arm. Neurologic examination shows ulnar nerve injury. It is expected that, distal to the injury, the axon and myelin will degenerate. Which of the following central cellular reactions is most likely to arise early in this patient?
- A. Wallerian degeneration
 - B. Apoptosis
 - C. Transneuronal degeneration
 - D. Chromatolysis
 - E. Microglial activation
11. A 40-year-old man develops progressive ascending muscle weakness and areflexia following a viral infection. Lumbar puncture reveals

elevated CSF protein with normal cell count. Electrodiagnostic studies show demyelination of peripheral nerves. Which of the following cell types is most likely affected in this patient?

- A. Oligodendrocytes
- B. Astrocytes
- C. Schwann cells
- D. Microglia
- E. Ependymal cells

12. A 60-year-old man comes to the physician because of a 6-month history of headaches and nausea with recurrent seizures. MRI of the head shows a tumor of the brain. Laboratory studies show that the tumor is primarily composed of cells forming extensive and abnormal myelin-like structures. Which of the following populations of glia cells are most likely implicated in the tumor in this patient?

- A. Astrocytes
- B. Oligodendrocytes
- C. Ependymal cells
- D. Schwann cells
- E. Microglia

NB 06 – Transport within the CNS

1. A 4-year-old boy is brought to the emergency department because of a 1-day history of fever, headache, and neck stiffness. A diagnosis of bacterial meningitis is made following a lumbar puncture for CSF analysis. Which of the following is the most appropriate site for a lumbar puncture in this patient?

- A. L1/L2
- B. L2/L3
- C. L3/L4
- D. L4/L5

2. A 67-year-old woman is brought to the physician because of memory problems that started about 3 months ago. Following assessment, a diagnosis of Alzheimer's disease is made. Imaging shows hydrocephalus. Which of the following is most likely present in this patient?
- A. Communicating hydrocephalus
 - B. Non-communicating hydrocephalus
 - C. Obstructive hydrocephalus
 - D. Hydrocephalus ex vacuo
 - E. Normal pressure hydrocephalus
3. A 1-week-old child is brought to the physician because of a rapid increase in the size of the head since birth. Examination and imaging studies are consistent with hydrocephalus due to aqueductal stenosis. Which of the following is most likely present in this patient?
- A. Communicating hydrocephalus
 - B. Non-communicating hydrocephalus
 - C. Hydrocephalus ex vacuo
 - D. Normal pressure hydrocephalus
4. A 74-year-old man is brought to the emergency department because of a 30-minute history of severe nausea and vomiting. Examination and investigation findings are consistent with the involvement of a circumventricular organ. Which of the following is the most likely involved in this patient's presentation?
- A. Amygdala
 - B. Area postrema
 - C. Basal ganglia
 - D. Cerebellum
 - E. Hippocampus

5. A 1-month-old child is brought to the physician because of an enlarged head, poor feeding, and irritability. A diagnosis of hydrocephalus is made following imaging studies, which show enlarged lateral and third ventricles, with a normal-sized fourth ventricle. Which of the following is most likely responsible for the findings in this patient?
- A. Brain tissue atrophy
 - B. Impaired CSF absorption
 - C. Obstruction of the cerebral aqueduct
 - D. Obstruction of the intraventricular foramen
6. A 6-month-old child is brought to the clinic with progressive head enlargement and irritability. Examination reveals bulging fontanelles. MRI shows dilation of all ventricles without evidence of obstruction. The physician suspects impaired cerebrospinal fluid absorption. Which structure is most likely dysfunctional?
- A. Choroid plexus of the lateral ventricles
 - B. Foramina of Monro (interventricular foramina)
 - C. Cerebral aqueduct (of Sylvius)
 - D. Arachnoid villi in the superior sagittal sinus
 - E. Subarachnoid cisterns
7. A 4-year-old child is brought to the emergency department with signs of increased intracranial pressure: headache, vomiting, and papilledema. CT imaging reveals overproduction of CSF due to a tumor. Which site is most likely the origin of this tumor?
- A. Ependymal lining of the cerebral aqueduct
 - B. Choroid plexus of the lateral ventricles
 - C. Arachnoid villi in the superior sagittal sinus
 - D. Astrocytes in the subdural space

NB 07 – Stimulated Excitable Cells

1. A 23-year-old woman comes to the physician because of a 3-week history of tingling affecting her left leg. Imaging studies show a central inflammatory demyelinating lesion is found in an ascending

somatosensory pathway. She receives a diagnosis of multiple sclerosis. What are the functional effects of the lesion on the electrical properties of the affected axons?

- A. Decreased membrane resistance and decreased length constant
 - B. Decreased membrane resistance and increased length constant
 - C. Increased membrane resistance and decreased length constant
 - D. Increased membrane resistance and increased length constant
2. A 64-year-old woman comes to the physician because of a 1-week history of weakness and numbness in her legs. Nerve conduction studies show reduced conduction velocity in her peripheral nerves. Which of the following is most likely responsible for the nerve conduction study findings in this patient?
- A. Increased axonal diameter
 - B. Increased myelination
 - C. Decreased myelination
 - D. Decreased intracellular resistance
 - E. Increased cross-sectional area
3. A 25-year-old man comes to the physician because of dizziness and palpitations that have lasted for 4 days. Examination and ECG findings are consistent with Long QT syndrome. A defect in the S4 transmembrane domain is implicated. Which of the following is a feature of the affected transmembrane domain?
- A. Negatively charged
 - B. Non-helical structure
 - C. Confers voltage sensitivity
 - D. Induces channel closing
 - E. Inhibits sodium flow
4. A 40-year-old man comes to the physician because of a 5-day history of knee pain and trouble straightening his left leg. During surgery for the removal of a ruptured medial meniscus, a drug is delivered into the lumbar subarachnoid space. Anesthesia and paralysis of the legs arise.

Which of the following structures is the most likely site of action of the drug?

- A. Acetylcholine receptors
- B. GABA receptors
- C. Glutamate receptors
- D. Glycine receptors
- E. Voltage-gated sodium channels
- F. Voltage-gated potassium channels
- G. Voltage-gated calcium channels
- H. Voltage-gated chloride channels

5. A 23-year-old man is brought to the emergency department because of a cut he sustained on his forearm while at work. Lidocaine is administered before the wound is sutured. Which of the following best explains the mechanism of action of the administered local anesthetic?

- A. Calcium channel block
- B. Increase in action potential
- C. Potassium channel block
- D. Sodium channel block
- E. Stimulation of calcium channels

6. A 53-year-old man comes to the physician because of a 36-hr history of headaches. Neurologic examination shows normal somatosensory function. When the physician applies pressure to the skin of the patient, somatosensory endings depolarize in a graded fashion. Which of the following electrical potentials is most likely under consideration in this patient?

- A. Action
- B. Excitatory postsynaptic
- C. Inhibitory postsynaptic
- D. Receptor
- E. Resting membrane

7. A 41-year-old woman is brought to the emergency department because of a 30-minute history of envenomation. Based on the description of the

envenomating organism, it is assumed that a toxin is blocking acetylcholine receptors at the neuromuscular junction. Which of the following events is most likely directly impacted by the toxin in this patient?

- A. Activation of voltage-gated sodium channels
- B. Activation of voltage-gated potassium channels
- C. Activation of ligand-gated sodium channels
- D. Activation of ligand-gated potassium channels
- E. Inactivation of voltage-gated sodium channels
- F. Inactivation of voltage-gated potassium channels

8. A research study investigates a genetic mutation that causes an abnormally leaky phospholipid bilayer. Such a defect would most directly impact the passive spread of electrical signals, leading to their rapid dissipation from the site of origin. Which of the following changes to cellular parameters is most likely in this context?

- A. Increased membrane resistance
- B. Decreased membrane resistance
- C. Increased intracellular resistance
- D. Decreased intracellular resistance

9. In an experimental setting, researchers modify neuronal properties such that larger excitatory postsynaptic potentials are required to provoke action potentials. Which of the following sites is most likely being manipulated by the researchers?

- A. Axonal terminal
- B. Postsynaptic receptor
- C. Myelin
- D. Axon hillock
- E. Synaptic vesicle

10. A 6-year-old boy is brought to the physician because of a 2-month history of muscle weakness and periodic paralysis. Genetic testing reveals a

mutation that causes voltage-gated sodium channels to inactivate slightly more slowly than normal. Which of the following electrical phenomena is most likely prolonged in this patient?

- A. Repolarization
- B. Hyperpolarization
- C. Depolarization
- D. Resting state

11. A 22-year-old woman comes to the physician because of a 48-hour history of progressive weakness. The underlying pathology relates to disrupted segments of axonal insulation. Which of the following cellular components is most likely under attack in this patient?

- A. Voltage-gated potassium channels
- B. Ligand-gated channels
- C. Nodes of Ranvier
- D. Voltage-gated sodium channels
- E. Myelin sheaths

12. In an experimental setting, the propagation of action potentials along axons with different properties is compared. The first axon is large and myelinated, whereas the second axon is small and unmyelinated. Which of the following descriptions best applies to the comparison of the propagating potentials reported in this study?

- A. Graded vs. all-or-none
- B. Saltatory vs. continuous
- C. Depolarization vs. repolarization
- D. Orthodromic vs. antidromic
- E. Inhibitory vs. excitatory

13. A 2-year-old boy is brought to the physician because of a 2-month history of reduced mobility and muscle stiffness. He receives a diagnosis of a rare genetic disorder characterized by hyperexcitable lower motor neurons. Electrophysiological studies show that these neurons fire action potentials in response to very weak stimuli that would normally be

subthreshold. Which of the following phenomena is most likely in the lower motor neurons of this patient?

- A. Tonic depolarization
- B. Increased action potential amplitude
- C. Elevated threshold for excitation
- D. Increased conduction velocity
- E. Decreased membrane resistance

14. A 30-year-old woman comes to the physician because of a 2-week history of weakness, particularly affecting her lower limbs. Laboratory studies show diffuse peripheral axonal degeneration with a corresponding loss of axonal diameter. Which of the following changes is most likely evident in this patient?

- A. Increased intracellular resistance
- B. Decreased intracellular resistance
- C. Increased membrane resistance
- D. Decreased membrane resistance

15. In an experimental setting, a novel neurotoxin binds to the inactivation gate of voltage-gated sodium channels in cultured neurons. Which of the following changes to the dynamics of action potentials is most likely in this experimental preparation?

- A. Enhanced repolarization
- B. Enhanced hyperpolarization
- C. Sustained depolarization
- D. Accelerated depolarization
- E. Reduced action potential amplitude

16. In an experimental setting, radiolabeled antibodies against voltage-gated sodium channels are applied to cultured neurons. Which of the following structures are most likely targeted?

- A. Axon terminals
- B. Nodes of Ranvier
- C. Myelin segments
- D. Somata
- E. Dendrites

NB 08 – Synaptic Transmission

1. A neuroscience professor conducting a study on synapses investigates the features of electrical synapses. Which of the following is the most likely feature of the synapse of interest?
 - A. Gap junctions
 - B. Slow intercellular communication
 - C. Indirect intercellular movement of ions
 - D. Communication between cells is always unidirectional

2. In an experimental setting, the quantal release of neurotransmitters is studied. Which of the following best describes the quantal content?
 - A. Content of synaptic vesicles
 - B. Transmitter released from a single vesicle
 - C. Number of vesicles per impulse
 - D. Smallest postsynaptic response
 - E. Number of impulses for a single vesicle

3. A 37-year-old man comes to the emergency department because of a 4-hour history of traumatic injury to the foot. Emergency surgery is prescribed. An anesthetist administers an acetylcholine receptor blocker to induce paralysis. Which of the following changes in synaptic transmission is most likely?
 - A. Reduced quantal content
 - B. Reduced quantal size
 - C. Increased quantal content
 - D. Increased quantal size

4. A neurotoxicologist researching the effects of various toxins on neurotransmission isolates a toxin that inhibits nicotinic acetylcholine receptors in the post-synaptic membrane. Which of the following is the most likely toxin studied?

- A. Botulinum toxin
- B. W-conotoxin
- C. Tetanus toxin
- D. Alpha bungarotoxin
- E. Beta bungarotoxin

5. A 58-year-old woman with a 5-month history of cervical dystonia (neck muscle spasms) comes to the physician for a follow-up. She is administered Myobloc (botulinum toxin type B). Which of the following is the most likely effect of this agent?

- A. Blockade of acetylcholine receptors
- B. Blockade of calcium channels
- C. Blockade of sodium channels
- D. Inhibition of acetylcholine release
- E. Inhibition of acetylcholinesterase

6. A 45-year-old man comes to the physician because of a 3-month history of muscle weakness and difficulty climbing stairs. Electromyography shows an increasing response to repetitive nerve stimulation. The patient is diagnosed with Lambert-Eaton myasthenic syndrome, which is caused by an immune attack on voltage-gated calcium channels. Which of the following phenomena best explains the weakness in this patient?

- A. Decreased quantal size
- B. Slower presynaptic repolarization
- C. Increased probability of vesicle fusion
- D. Reduced numbers of synaptic vesicles released per action potential
- E. Prolonged postsynaptic depolarization

7. A 28-year-old man comes to the physician because of a 1-day history of blurred vision and difficulty swallowing. Physical examination shows that he has bilateral ptosis and generalized muscle weakness. He reports recently eating home-canned vegetables. It is suspected that a food-borne toxin is cleaving a protein involved in vesicular fusion. Which of the following transmission-related processes is most likely impaired in this patient?

- A. Presynaptic synthesis
- B. Exocytosis
- C. Reuptake of choline
- D. Postsynaptic binding
- E. Opening of voltage-gated calcium channels

8. In an experimental setting involving cultured neurons, a ligand-gated receptor is mutated such that it functions as a metabotropic receptor (as opposed to an ionotropic receptor). Which of the following descriptions best applies to the altered pattern of neurotransmission in this experimental preparation?

- A. More rapid and transient
- B. G-protein-mediated
- C. Reuptake-independent
- D. Direct activation of ion channels
- E. Synaptic metabolism of transmitter

9. A 30-year-old woman comes to the physician because of the need for management of a mood disorder that is being treated with a serotonin-selective reuptake inhibitor (SSRI). Which of the following outcomes related to serotonergic transmission arises consequent to the administration of the medication in this patient?

- A. Increased synthesis
- B. Suppressed degradation
- C. Increased synaptic longevity
- D. Sensitization of receptors
- E. Enhanced release

10. In an experimental setting involving cultured neurons, a defective mutation of the vesicle-associated transporter for dopamine is under study. Transport of which of the following ions is most likely impaired in these neurons?

- A. Sodium
- B. Potassium
- C. Chloride

- D. Calcium
- E. Hydrogen

11. A 22-year-old woman comes to the physician because of a 2-hour history of intense anxiety. The prescribed medication enhances the GABA-dependent influx of chloride, thereby reducing cellular excitability. Which of the following structures is most likely targeted by the drug?
- A. Metabotropic receptors
 - B. Ionotropic receptors
 - C. Voltage-gated potassium channels
 - D. Voltage-gated sodium channels
 - E. Acetylcholine receptors
12. A 6-month-old infant is brought to the physician because of a 1-day history of muscle weakness and poor feeding. The parents say that they recently changed to honey-sweetened pacifiers. Botulinum toxemia is suspected. Which of the following outcomes involving the neuromuscular junction is expected in this patient?
- A. Decreased quantal content
 - B. Diminished degradation of synaptic acetylcholine
 - C. Increased frequency of miniature end-plate potentials
 - D. Increased postsynaptic receptor density
 - E. Presynaptic hyperpolarization
13. A 70-year-old man is brought to the physician because of a 6-month history of memory loss and cognitive decline. Alzheimer's disease is suspected. An inhibitor of acetylcholinesterase is prescribed. Which of the following effects of the drug is most likely in this patient?
- A. Enhanced synthesis of acetylcholine
 - B. Inhibited reuptake of choline
 - C. Proliferation of postsynaptic acetylcholine receptors
 - D. Prolongation of the synaptic longevity of acetylcholine
 - E. Enhanced presynaptic calcium influx

14. In an experimental setting involving cultured neurons, a genetic mutation causes dysfunction of a presynaptic protein involved in the docking of synaptic vesicles. This defect impairs the release of neurotransmitter. Which of the following outcomes is most likely in this experimental study?
- A. Increased amplitude of the postsynaptic potential
 - B. Lowered threshold for the firing of action potentials
 - C. Prevention of the postsynaptic potential
 - D. Prolongation of the rising phase of the action potential
 - E. Reduced amplitude of the postsynaptic potential
15. A 16-year-old girl is brought to the physician because of a 6-month history of weakness. Laboratory studies show a genetic mutation affecting the enzyme that metabolizes the precursor molecule. Which of the following steps in the life cycle of the neurotransmitter is most likely affected in this patient?
- A. Reuptake
 - B. Vesicular packaging
 - C. Degradation
 - D. Synthesis
 - E. Retrograde transport

Questions with Answers and Explanations

NB 01 – Neuroembryology

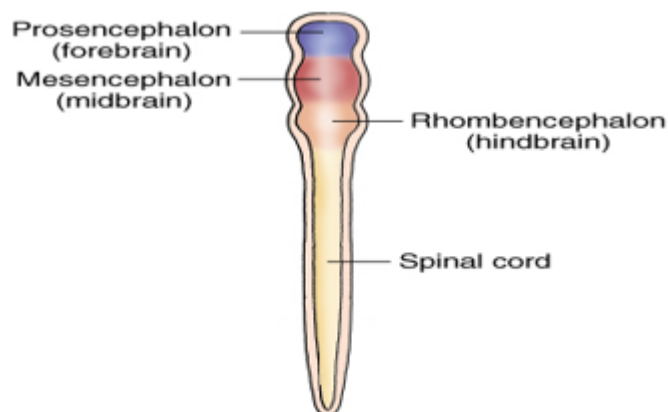
1. A 36-year-old woman in her fourth week of pregnancy comes to the physician to enquire about her baby's development. The development of the nervous system is explained to her. The development of which of the following is most likely induced by the notochord?
- A. Neural tube

- B. Neural fold
- C. Neural plate
- D. Neural crest
- E. Neural sheath

Answer Key: C

The notochord causes differentiation of the ectoderm found above it to neuroectoderm, thus inducing neural plate development. The neural fold occurs when the lateral edges come together, and the neural tube results from the neural folds fusing. Neural crest are the cells from the lateral margins of the neural groove that separate and migrate dorsally. The neural sheath is a layer of myelin and connective tissue that surrounds and insulates nerve fibers.

2. The figure below shows the primary vesicles of the nervous system. From which of the following would the epithalamus most likely develop?



- A. Telencephalon
- B. Diencephalon
- C. Mesencephalon
- D. Myelencephalon
- E. Metencephalon

Answer Key: B

The prosencephalon forms the telencephalon and diencephalon. The telencephalon forms the cerebrum. The diencephalon forms the thalamus, hypothalamus, and epithalamus. The mesencephalon is the midbrain. The metencephalon and myelencephalon come from the rhombencephalon and form the pons, cerebellum, and medulla.

3. In the development of the forebrain, which of the following structures most likely develops from the telencephalon?

- A. Mammillary bodies
- B. Lateral ventricle
- C. Anterior pituitary
- D. Massa Intermedia
- E. Third ventricle

Answer Key: B

Answers A, C, D, and E are all structures that develop from the diencephalon. The lateral ventricle (B) is the only response that develops from the telencephalon forming within each cerebral hemisphere.

4. Cranial nerve nuclei situated more laterally in the brainstem develop from the _____ and are most likely _____ neurons.

- A. Floor plate, Sensory
- B. Basal plate, Sensory
- C. Basal plate, Motor
- D. Alar Plate, Sensory
- E. Alar Plate, Motor

Answer Key: D

In the morphogenesis of the brainstem the neural canal expands to become the 4th ventricle, the roof plate expands, and the alar plate comes to lie lateral to the basal plate. The alar plate is sensory, and the structures associated with it are also sensory. The basal plate and its associated derivatives are motor and are found medially. Thus, the motor cranial nerves are medial to the sensory cranial nerves.

5. A newborn is being evaluated for congenital malformations of the brain. Imaging studies show a cystic enlargement of the fourth ventricle and an absence of the cerebellar vermis. Which of the following is the most likely condition in this patient?

- A. Anencephaly
- B. Syringomyelia
- C. Spina bifida
- D. Dandy-Walker Syndrome
- E. Encephalocele

Answer Key: D

Dandy-Walker Syndrome is characterized by a congenital absence of the foramen of Luschka and the foramen of Magendie. There is also complete/partial agenesis of the cerebellar vermis and cystic dilation of the posterior fossa communicating with the fourth ventricle. Anencephaly – partial or complete absence of the brain. Syringomyelia - A cavity formed in the region of the central canal filled with CSF. Spina bifida – failure of vertebral canal closure. Encephalocele - failure of portions of the anterior neural tube to close.

6. A 23-year-old primigravida at 3 weeks of gestation learns that her embryo is undergoing neural tube development. As development progresses, the brainstem walls will reorganize so that sensory nuclei shift laterally while motor nuclei remain medial. This rearrangement most likely results from which developmental process?

- A. Formation of the cervical flexure
- B. Formation of the pontine flexure
- C. Migration of neural crest cells into the hindbrain
- D. Rotation of the mesencephalon around the cephalic flexure
- E. Expansion of the cerebral hemispheres and compression of the midbrain

Answer Key: B

The pontine flexure (between metencephalon and myelencephalon) causes the dorsal wall of the brainstem to spread apart. This lateralizes the alar plate (sensory nuclei) and leaves the basal plate (motor nuclei) medial — a hallmark of brainstem organization.

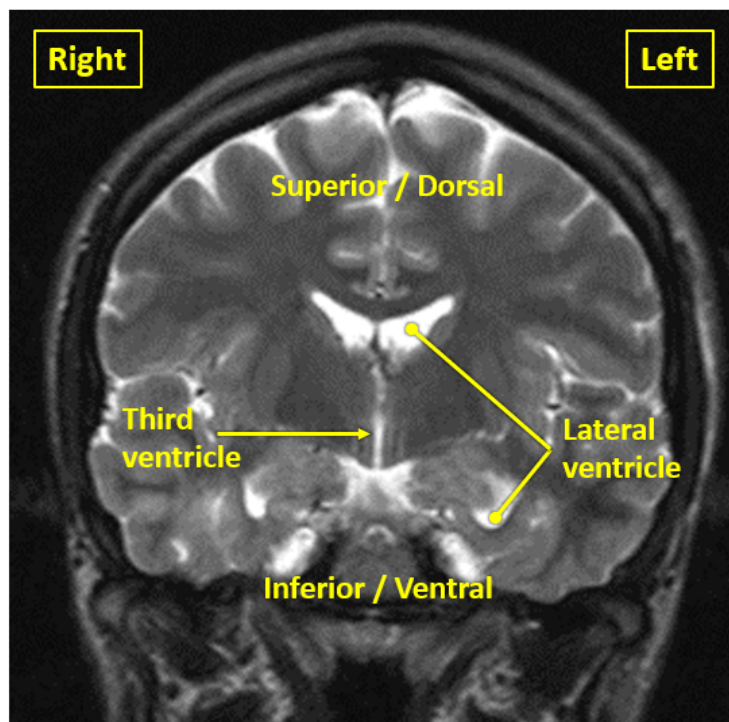
NB 02 – General Morphology of the Nervous System

1. A 1-week-old child is brought to the physician because of a rapid increase in head size since birth. Examination and imaging studies are consistent with hydrocephalus due to aqueductal stenosis (narrowing of the cerebral aqueduct). The affected structure is most likely associated with which of the following?
- A. Diencephalon
 - B. Mesencephalon
 - C. Metencephalon
 - D. Myelencephalon
 - E. Telencephalon

Answer Key: B

The cerebral aqueduct connects the third and fourth ventricles and is closely related to the midbrain - mesencephalon.

2. A 50-year-old woman comes to the physician because of a headache that has lasted for 3 days. Vital signs are within normal limits. Examination findings show no abnormality. Imaging studies are subsequently done for the patient. Which of the following is the most likely orientation of the image below?



- A. Sagittal
- B. Transverse
- C. Horizontal
- D. Coronal
- E. Axial

Answer Key: D

The image is a coronal section of the brain. A sagittal plane divides into left and right parts, while a horizontal plane (also known as axial and transverse) divides into superior and inferior parts.

3. A 72-year-old man is brought to the physician because of difficulty in understanding spoken language that has lasted for 2 days. Examination findings and imaging are consistent with sensory aphasia secondary to a stroke. The affected lobe lies inferior to the lateral sulcus. Which of the following lobes is most likely affected in this patient?

- A. Parietal
- B. Frontal

- C. Occipital
- D. Insular
- E. Temporal

Answer Key: E

Sensory aphasia is also known as Wernicke's aphasia or receptive aphasia. Wernicke's area is in the temporal lobe, which lies inferior to the lateral sulcus.

4. A 57-year-old man comes to the physician because of a 2-week history of tingling sensations on the left side of his body. Neurological examination shows general somatosensory loss affecting the left side of the face and the left hand. Which of the following lobes is most likely lesioned?

- A. Left frontal
- B. Left temporal
- C. Left occipital
- D. Right parietal
- E. Right occipital
- F. Right frontal
- G. Left parietal
- H. Right temporal

Answer Key: D

The right parietal lobe mediates somatosensation from the left side of the body.

5. A 41-year-old woman comes to the physician because of a 12-hour history of clumsiness. Strength appears to be normal for all limbs. However, she lacks coordination with the right arm and leg. Imaging studies show a lesion in a structure that arises developmentally from the dorsal aspect of the pons. Which of the following structures is most likely injured?

- A. Spinal cord

- B. Medulla
- C. Midbrain
- D. Thalamus
- E. Cerebellum

Answer Key: E

The cerebellum arises developmentally from the pons and is thus part of the metencephalon. The cerebellum is also responsible.

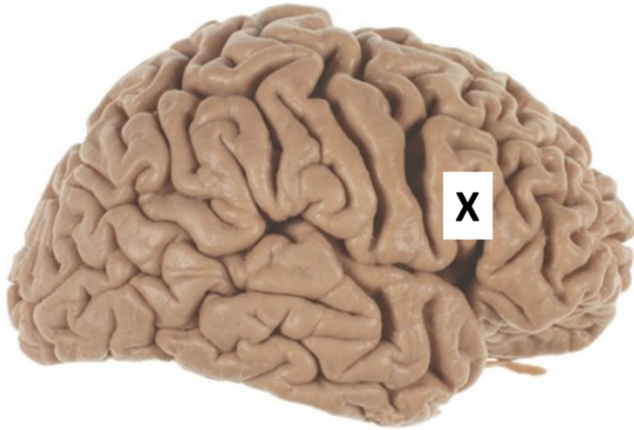
6. A 44-year-old man is brought to the emergency department because of a 6-day history of worsening headaches, nausea, and vomiting. Imaging shows dilation of the third ventricle and both lateral ventricles. Which of the following ventricular passages is most likely blocked?

- A. Right interventricular foramen
- B. Left interventricular foramen
- C. Interventricular foramina
- D. Cerebral aqueduct
- E. Foramen of Magendie
- F. Foramina of Luschka

Answer Key: D

The drainage of the enlarged ventricles depends most directly upon the cerebral aqueduct.

7. A 6-year-old boy is brought to the physician by his mother because of a 2-month history of headaches and difficulty concentrating. The headaches are localized to the right side of the head. MRI of the brain shows a tumor residing at the location indicated by the 'X' in the image of a brain specimen below. If the tumor growth progresses ventrally, which of the following structures will most likely be affected?



[Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 11e](#) Figure 2-11, modified

- A. Supramarginal gyrus
- B. Middle frontal gyrus
- C. Postcentral gyrus
- D. Precentral gyrus
- E. Superior temporal gyrus

Answer Key: E

The tumor is in the inferior frontal gyrus. The superior temporal gyrus is ventral (or inferior) to the tumor and hence, could be affected if tumor growth continued in this direction. The supramarginal, precentral, and postcentral gyri are all posterior/caudal to the lesion site, while the middle frontal gyrus is dorsal/superior.

8. A 51-year-old woman comes to the physician because of a 2-day history of difficulty looking upwards voluntarily. Imaging studies show a lesion affecting the superior colliculi. Which of the following brain regions are the affected structures most likely located?
- A. Ventral midbrain
 - B. Dorsal midbrain
 - C. Ventral pons
 - D. Dorsal pons
 - E. Ventral medulla

Answer Key: B

The superior colliculi are located in the tectum of the midbrain (dorsal midbrain) and are primarily involved in visual reflexes and control of vertical gaze. The inferior colliculi are also located in the dorsal midbrain, where they serve as major relay centers in the auditory pathway.

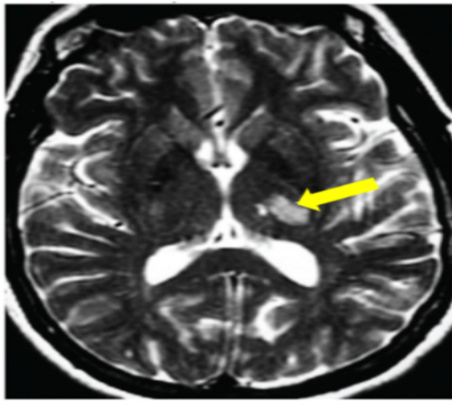
9. A 39-year-old man is brought to the emergency department because of a 3-hour history of loss of sensation on the right side of his body. Neurologic examination shows loss of touch and vibration sensations on the right side of his body. Imaging studies show a left diencephalic lesion. Which of the following diencephalic structures is most likely affected in this patient?

- A. Epithalamus
- B. Hypothalamus
- C. Subthalamus
- D. Thalamus

Answer Key: D

The thalamus is a diencephalic structure responsible for relaying sensory information to the sensory cortical areas. The sensory information from the right side of the body goes through the left thalamus. The epithalamus regulates the circadian rhythm by producing melatonin, while the hypothalamus plays a role in various homeostatic and neuroendocrine functions. The subthalamus is involved in the regulation of movement.

10. A 56-year-old woman is brought to the emergency department by her son because of a 4-hour history of numbness. The woman is drowsy on arrival but is oriented to time, place, and person. She suffered sudden onset of headache followed by numbness in her left hand. She has a 30-year history of hypertension and does not regularly take her medication. She admits regular, heavy consumption of alcohol. T2-weighted MRI of the brain shows a unilateral hyperintensity indicative of an intracerebral haemorrhage in the posterior diencephalon, indicated by the yellow arrow in the attached image. With which of the following components of the ventricular system is the affected structure most likely associated?



Haines, Neuroanatomy Atlas, © Wolters Kluwer, Figure 2-44H

- A. Left lateral ventricle
- B. Right lateral ventricle
- C. Cerebral aqueduct
- D. Third ventricle
- E. Fourth ventricle

Answer Key: D

The diencephalon is associated with the midline third ventricle.

11. A 17-year-old boy is brought to the emergency department having hit his head while colliding with another player during a football game. He had a brief loss of consciousness immediately following the incident. He is alert and oriented on arrival. Neurological examination is normal. CT scan of the head does not show any evidence of intracranial bleeding. However, an incidental finding of a neoplasm deriving from a midline diencephalic structure that forms part of the posterior wall of the third ventricle is made. In which of the following structures is the neoplasm most likely located?

- A. Thalamus
- B. Pituitary gland
- C. Pineal gland
- D. Mammillary bodies
- E. Hypothalamus

Answer Key: C

The pineal gland is a midline structure of the diencephalon that forms part of the posterior wall of the third ventricle. The thalamus forms the lateral wall of the third ventricle. The hypothalamus also forms part of the lateral

wall. The pituitary gland and the mammillary bodies are inferior to the hypothalamus.

NB 03 – Organisation of the Nervous System

1. A 64-year-old man comes to the physician because of a 6-hour history of weakness affecting the left hand. Neuroimaging studies show a small cortical lesion within the frontal lobe. Which of the following structures most closely overlaps with the site of the injury?

- A. Motor nucleus
- B. Motor tract
- C. Somatosensory nucleus
- D. Somatosensory tract

Answer Key: A

The cortex is a gray nucleated structure, substantially occupied by cell bodies. The weakness reflects the disruption of a motor system.

2. A 25-year-old woman is brought to the physician because of an inability to pronounce words correctly for 3 weeks. On examination, she is unable to repeat words or sentences. Imaging studies show damage to an association fiber, and she is subsequently diagnosed with conduction aphasia. Which of the following fibers is most likely damaged in this patient?

- A. Internal capsule
- B. Corona radiata
- C. Corpus callosum
- D. Arcuate fasciculus
- E. Anterior commissure

Answer Key: D

Conduction aphasia results from damage to the arcuate fasciculus, which is an association fiber. The internal capsule and corona radiata are

projection fibers, while the corpus callosum and anterior commissure are commissural fibers.

3. A 20-year-old man is brought to the emergency room following a car crash about 20 minutes ago. He is unconscious. Imaging shows an epidural hematoma, with herniation of the uncus and compression of the midbrain. Which of the following cranial nerves is most likely affected?

- A. Olfactory
- B. Optic
- C. Oculomotor
- D. Facial
- E. Vestibulocochlear

Answer Key: C

The oculomotor nerve is most likely affected as it arises from the midbrain. The olfactory and optic nerves arise from the telencephalon and diencephalon respectively. While the facial and vestibulocochlear arise from the pontomedullary junction.

4. A 32-year-old man is brought to the emergency room by the paramedics following a motor vehicle accident. Examination and investigation findings were consistent with a right hemisection (damage to the right side) of the spinal cord at the C5 vertebral level. Which of the following neurological deficits would this patient most likely have?

- A. Paralysis of both upper limbs
- B. Paralysis of both lower limbs
- C. Paralysis of the left upper limb
- D. Paralysis of the left lower limb
- E. Paralysis of the right lower limb

Answer Key: E

A right hemisection of the spinal cord at the C5 vertebral level (below the decussation of the motor pathway) will result in right-sided motor deficits. Paralysis of the right lower limb is therefore the best answer choice. This lesion can also cause paralysis of the right upper limb. The left upper and lower limbs would be spared.

5. A 50-year-old woman comes to the physician because of a headache and double vision that have lasted for 3 days. On neurologic examination, she is unable to move her right eye laterally. Which of the following cranial nerves is most likely affected in this patient?

- A. Olfactory
- B. Optic
- C. Abducens
- D. Facial
- E. Vestibulocochlear

Answer Key: C

Since the right eye cannot move laterally, a cranial nerve responsible for eye movements is affected. Regarding eye movements, the oculomotor, trochlear, and abducens nerves are responsible.

6. A 53-year-old man comes to the physician because of a 3-month history of memory problems. Imaging studies show a unilateral hippocampal lesion. Which of the following lobes of the cerebrum is most likely affected in this patient?

- A. Frontal
- B. Parietal
- C. Insular
- D. Occipital
- E. Temporal

Answer Key: E

The hippocampus is located in the medial temporal lobe and plays a central role in learning and memory consolidation. Lesions of the hippocampus can cause memory impairment.

7. A 64-year-old woman comes to the physician because of a 1-month history of double vision. Imaging studies show a dorsal brainstem lesion. Which of the following cranial nerves is most likely affected in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V
- F. VI
- G. VII
- H. VIII
- I. IX
- J. X
- K. XI
- L. XII

Answer Key: D

Cranial nerve IV (trochlear nerve) is the only cranial nerve that emerges from the dorsal brainstem, specifically the midbrain.

8. A 64-year-old man comes to the physician because of a 2-week history of headaches. MRI of the head shows a tumor located laterally at the right ponto-medullary junction. Which of the following cranial nerves is most likely affected in this patient?

- A. Olfactory
- B. Optic
- C. Oculomotor

- D. Trochlear
- E. Trigeminal
- F. Abducens
- G. Facial
- H. Vestibulocochlear
- I. Glossopharyngeal
- J. Vagus
- K. Accessory
- L. Hypoglossal

Answer Key: H

The vestibulocochlear nerve, responsible for hearing and balance, arises from the lateral pontomedullary junction. Two other nerves arise from the pontomedullary junction: the abducens–medial and the facial–intermediate.

NB 04 – Meninges and Arteries

1. A 13-year-old girl is brought to the emergency department by her parents because of a 12-hour history of rising temperature, photophobia, a stiff neck, and irritability. Laboratory studies involve the drawing of CSF. Between which of the following tissue layers is the sample most likely drawn?
 - A. Skull and periosteal dura
 - B. Periosteal dura and meningeal dura
 - C. Meningeal dura and arachnoid
 - D. Arachnoid and pia
 - E. Pia and spinal cord

Answer Key: D

There is a possibility of meningitis in this patient. CSF analysis is therefore important for diagnosis. The subarachnoid space may furnish a sample of CSF for confirmatory laboratory studies. The subarachnoid space is bounded by the arachnoid layer and the pia.

2. A 36-year-old man is brought to the emergency department by paramedics because of a 1-hour history of trauma to the left side of the head. He lost consciousness at the moment of impact but regained consciousness within 5 minutes. The skin on the left side of his head is split. Six hours later, he becomes lethargic and shows signs of cerebral compression from an accumulating mass of blood. Which of the following arteries is most likely ruptured?

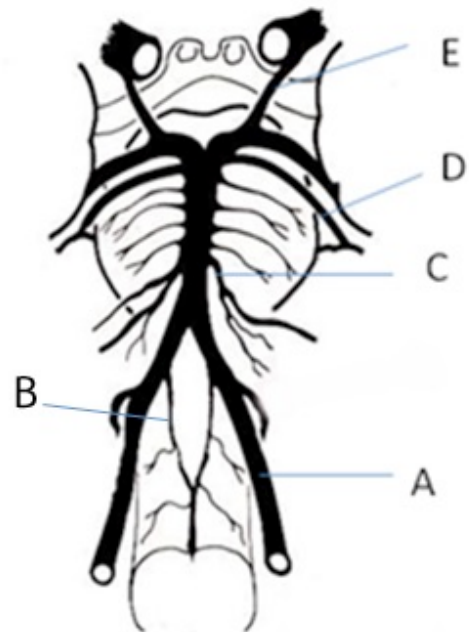
- A. Middle cerebral
- B. Posterior cerebral
- C. Middle meningeal
- D. Anterior cerebral
- E. Internal carotid
- F. Vertebral
- G. Basilar

Answer Key: C

The history is consistent with an epidural hemorrhage, which commonly arises with a skull fracture. The injury to the side of the head implicated the middle meningeal artery.

3. A 55-year-old man is brought to the emergency department because of a 2-hour history of loss of balance, nausea, vomiting, and a sensation that the room is spinning. A lesion situated at the junction of the medulla and pons is suspected. Which of the following subarachnoid arteries is most likely implicated in the injury (see attached image)?

- A. A
- B. B
- C. C
- D. D
- E. E



Answer Key: C

C indicates the anterior inferior cerebellar artery. This artery supplies Cranial Nerve VIII, including the vestibular portion, which is tied to equilibrium.

4. A 69-year-old woman comes to the physician because of a 2-week history of headache and confusion following minor head trauma. Imaging studies of the head show a right-sided crescent-shaped hemorrhage. Rupture of which of the following vessels is/are the most likely in this patient?

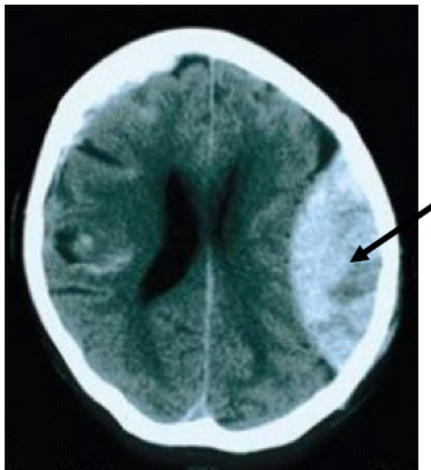
- A. Anterior communicating artery
- B. Anterior cerebral artery
- C. Bridging veins
- D. Lenticulostriate artery
- E. Middle meningeal artery

Answer Key: C

The findings in this case are consistent with a subdural hemorrhage, which arises from the rupture of bridging veins. The middle meningeal artery is implicated in epidural hemorrhage, while the lenticulostriate arteries are associated with an intracerebral hemorrhage. The anterior cerebral and

anterior communicating arteries are found in the subarachnoid space, and rupture of these vessels results in subarachnoid hemorrhage.

5. A 24-year-old man is brought to the emergency department following a motor vehicle accident 30 minutes ago. Imaging studies show an intracranial hemorrhage. Which of the following is the most likely hemorrhage depicted by the arrow (see attached image)?

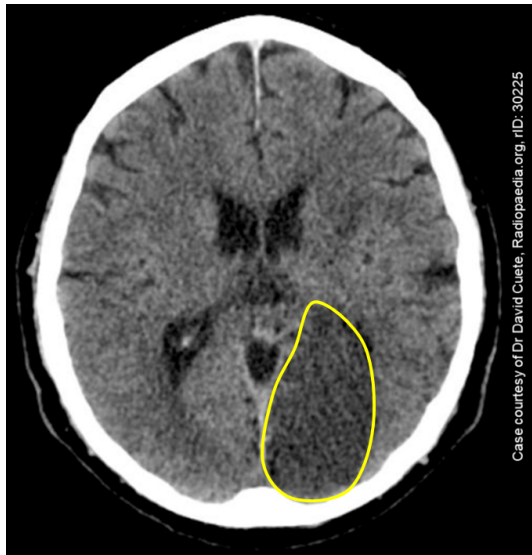


- A. Epidural
- B. Subdural
- C. Subarachnoid
- D. Intracerebral

Answer Key: A

In epidural hemorrhage, leaking blood separates the stiff periosteal dura from the skull, yielding a characteristic biconvex (lenticular; lens-shaped) mass adjacent to the fracture on imaging. For subdural hemorrhage, imaging shows a crescent-shaped subdural mass. Subarachnoid hemorrhage - the leaking blood assumes a tree-like appearance while following sulci and other contours of the subarachnoid space (a star-shaped appearance may also be noticed if blood accumulates in the circle of Willis). Intracerebral hemorrhage – there is bleeding within brain tissue or into the ventricles.

6. A 57-year-old woman is brought to the emergency department because of visual problems that began 8 hours previously. Neurologic examination shows right visual field deficits. CT scan of the head shows an infarction (outlined in yellow in the image). Identify the cerebral artery most likely occluded and the gyrus affected.



- A. Left anterior, posterior paracentral
- B. Left middle, postcentral
- C. Left posterior, lingual
- D. Right anterior, anterior paracentral
- E. Right middle precentral
- F. Right posterior, cuneus

Answer Key: C

The axial CT image shows a lesion affecting the left posterior, medial cortical structures. This region most likely receives blood supply from the left posterior cerebral artery (PCA). The lingual gyrus is a structure within the medial occipital lobe, which is indeed supplied by the PCA.

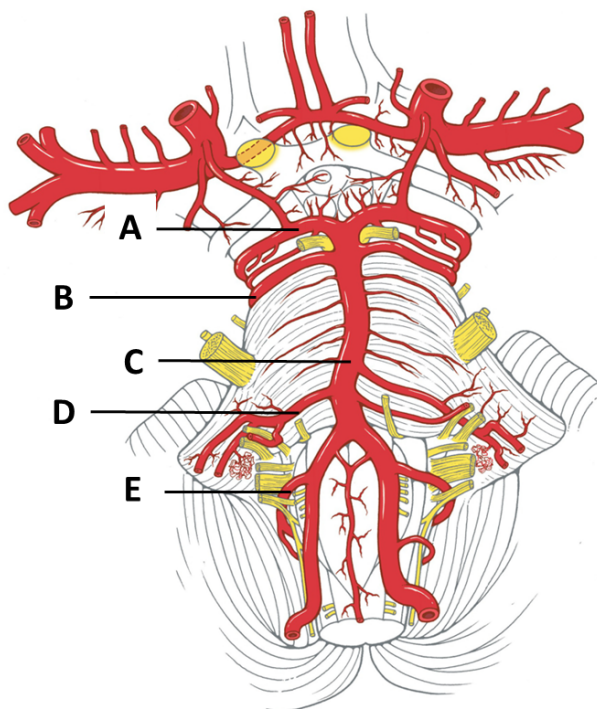
7. A 52-year-old man is brought to the emergency department because of dizziness and nausea that began 6 hours ago. Neurologic examination shows somatosensory and cranial nerve deficits that are consistent with a dorsolateral medullary lesion. Which of the following arteries is most likely occluded?

- A. Posterior cerebral
- B. Basilar
- C. Anterior spinal
- D. Posterior inferior cerebellar
- E. Superior cerebellar

Answer Key: D

Of the options given, only the anterior spinal (C) and posterior inferior cerebellar (D) arteries contribute to the supply of the medulla. The anterior spinal artery supplies medial medullary structures, while the posterior inferior cerebellar artery (PICA) branches from the rostral vertebral artery and travels dorsally and laterally to supply dorsolateral structures within the medulla.

8. A 68-year-old man is brought to the emergency department because of weakness that began 12 hours ago. Neurologic examination shows motor, somatosensory, and cranial nerve deficits that are consistent with a ventral midbrain lesion. Which of the following arteries is most likely occluded?



[Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 11e](#) Figure 2-21, modified

Answer Key: A

Vessel A is the posterior cerebral artery (PCA), branches of which supply structures in the ventral midbrain. Note that other branches of PCA supply cortical structures, such as the medial occipital lobe and parts of the inferior temporal lobe. You should be able to identify the other labelled vessels and predict the brainstem region they supply.

9. A 62-year-old man is brought to the emergency department because of a 1-hour history of a sudden-onset of severe headache and nuchal rigidity. CT scan of the head shows diffuse blood within the sulci and basal cisterns, outlining the convolutions of the brain. Which of the following vessels is/are most likely ruptured in this patient?

- A. Middle meningeal artery
- B. Lenticulostriate artery
- C. Bridging veins
- D. External carotid artery
- E. Posterior communicating artery

Answer Key: E

The sudden headache and nuchal rigidity with corresponding indications of blood in the subarachnoid space indicate a subarachnoid hemorrhage. A common cause of non-traumatic subarachnoid hemorrhage is a ruptured saccular (berry) aneurysm, and a significant percentage of these arise from the posterior communicating artery or anterior communicating artery, which are part of the circle of Willis and lie within the subarachnoid space.

- A. Middle meningeal artery: Injury to this artery typically involves trauma and causes an epidural hematoma, which presents as a biconvex/lenticular collection between the skull and dura, not diffuse blood in sulci.
- B. Lenticulostriate artery: Rupture of these deep penetrating arteries typically causes intraparenchymal hemorrhage (hypertensive hemorrhage), primarily affecting the basal ganglia and nearby white matter, not diffuse subarachnoid blood.
- C. Bridging veins: Tearing of bridging veins typically causes a subdural hematoma, which is crescent-shaped and located between the dura and arachnoid, often due to trauma.

D. External carotid artery: Ruptures of the external carotid artery would not lead to infiltration of blood into the subarachnoid space.

10. A 28-year-old man comes to the physician because of a 3-month history of nausea and headaches. MRA of the head shows an arteriovenous malformation involving major subarachnoid vessels supplying the insular cortex and lateral cerebral surface. Which of the following arteries is most likely implicated in this patient?

- A. Anterior cerebral
- B. Posterior cerebral
- C. Middle cerebral
- D. Basilar
- E. Vertebral

Answer Key: C

The Middle Cerebral Artery is the largest branch of the internal carotid artery and is a major component of the anterior circulation. It courses through the Sylvian fissure and gives off numerous branches that supply the lateral surface of the frontal, parietal, and temporal lobes, including the insular cortex. Therefore, a vascular anomaly in this region would primarily involve the middle cerebral artery and its branches.

A. Anterior cerebral: This vessel supplies the medial aspects of the frontal and parietal lobes.

B. Posterior cerebral: This artery supplies the occipital lobe, inferior temporal lobe, and parts of the thalamus/midbrain.

D. Basilar: This artery supplies the pons, while passing blood to the anterior inferior cerebellar, posterior inferior cerebellar, and posterior cerebral arteries.

E. Vertebral: The vertebral artery most directly supplies the ventral medulla.

11. A 58-year-old man is brought to the emergency department because of a 4-hour history of confusion and a loss of awareness of the left side of space. CT scan of the head shows a hemorrhage arising from a malformation of the middle cerebral artery that follows the cortical sulci of the right lateral parietal cortical surface. The partially clotted mass is not restricted to the anatomical boundaries of the skull and assumes

neither a lenticular nor crescent-like shape. Which of the following compartments is most likely filling with blood in this patient?

- A. Epidural space
- B. Subdural space
- C. Subarachnoid space
- D. Periventricular space
- E. Ventricular system

Answer Key: C

The description of the hemorrhagic mass best indicates a subarachnoid hemorrhage, perhaps arising from a lateral parietal arteriovenous malformation (note that the blood follows the sulci). A condition called left hemineglect syndrome also indicates right lateral parietal involvement.

A. Epidural space: This space is between the skull and dura. Hemorrhages here (epidural hematomas) are typically lenticular and do not follow the surface of the brain.

B. Subdural space: This space is between the dura and arachnoid. Hemorrhages here (subdural hematomas) are crescent-shaped and do not involve the brain parenchyma directly, although they can compress it.

D. Periventricular space: Blood can occupy this space following rupture of parenchymal vascular malformations. However, such lesions are irregular in shape and do not primarily follow cortical sulci.

E. Ventricular system: While intraparenchymal or interventricular arteriovenous malformations can rupture into the ventricles (intraventricular hemorrhage), the primary location described for the clotted mass is on the lateral surface of the brain, not near the ventricular system. There is no indication that the blood is filling the ventricles in this patient.

12. A 50-year-old man is brought to the emergency department because of a 1-hour history of a sudden headache. CT scan of the head confirms a subarachnoid hemorrhage. Which of the following layers prevents the blood from contacting the surface of the brain in this patient?

- A. Dura mater
- B. Arachnoid
- C. Pia mater
- D. Periosteal dura

E. Ligamentum denticulatum

Answer Key: C

The subarachnoid space is located between the arachnoid and the pia mater. The pia mater is the innermost meningeal layer, a delicate membrane that closely adheres to the surface of the brain and spinal cord, following all contours (gyri, sulci, and fissures). The major cerebral arteries and veins are situated within the subarachnoid space, superficial to the pia mater and thus separated from the surface of the brain.

A. Dura mater: The outermost and toughest meningeal layer is far removed from direct adhesion to the brain.

B. Arachnoid: This is the intermediate meningeal layer that sits between the dura and pia and forms the outer boundary of the subarachnoid space. It does not tightly adhere to the contours of the brain.

D. Periosteal dura: This is the outer layer of dura, and it is most intimately related to the skull.

E. Ligamentum denticulatum: These are pial extensions that anchor the spinal cord to the dura, but the pia itself is the layer covering the cord.

13. A 30-year-old woman comes to the physician because of a 3-hour history of visual problems. Neurologic examination shows a palsy related to dysfunction of the oculomotor nerve. Subsequent imaging reveals an aneurysm originating from the circle of Willis, specifically at the junction of the posterior communicating artery. Which of the following anatomical spaces would most likely fill with blood in the event of a rupture of the aneurysm in this patient?

A. Epidural space

B. Subdural space

C. Subarachnoid space

D. Intraparenchymal space

E. Cavernous sinus

Answer Key: C

The circle of Willis, including the posterior communicating artery, lies entirely within the subarachnoid space at the base of the brain. Aneurysms originating from these vessels, when ruptured, release blood directly into

the cerebrospinal fluid within the subarachnoid space, leading to a subarachnoid hemorrhage.

A. Epidural space: This virtual space is between the skull and dura and may fill with blood after meningeal arterial trauma but not a ruptured aneurysm.

B. Subdural space: This virtual space is between the dura and arachnoid and may fill with blood following rupture of bridging veins.

D. Intraparenchymal space: The typical location of an intracranial aneurysm is within the subarachnoid space and not within the parenchyma.

E. Cavernous sinus: While some aneurysms (e.g., internal carotid) can extend into the cavernous sinus, the clinical presentation in this patient is not entirely typical of intracavernous involvement of the internal carotid artery.

14. During a craniotomy, a neurosurgeon encounters the outermost, tough, fibrous layer of the cranial meninges. This layer forms two principal reflections or folds that partially separate parts of the brain. Which of the following structures is/are most likely under observation in this surgical patient?

A. Denticulate ligaments

B. Filum terminale

C. Falx cerebri

D. Arachnoid villi

E. Choroid plexus

Answer Key: C

The outermost tough fibrous layer of the cranial meninges is the dura mater. The dura mater forms several important folds that partition the cranial cavity, including the falx cerebri (which separates the cerebral hemispheres) and the tentorium cerebelli (which separates the cerebrum from the cerebellum).

A. Denticulate ligaments: These are pial extensions that anchor the spinal cord to the dura mater.

B. Filum terminale: This is an extension of the pia mater that anchors the spinal cord to the coccyx.

- D. Arachnoid villi: These are outgrowths of the arachnoid that project into the dural venous sinuses for CSF reabsorption.
- E. Choroid plexus: This is a vascular structure within the ventricles responsible for producing cerebrospinal fluid (CSF), and is not a dural fold.

15. A 42-year-old woman comes to the physician because of a 2-hour history of trauma to the side of her head. She does not know whether she was unconscious after the impact. CT of the head shows a fracture of the temporal bone and an associated bleed. Which of the following structures is most likely related to the damaged vessel in this patient?
- A. Pia mater
 - B. Arachnoid
 - C. Dura mater
 - D. Leptomeninges
 - E. Choroid plexus

Answer Key: C

An epidural hematoma occurs in the epidural space, which is the potential space between the inner surface of the skull and the outer (periosteal) layer of the dura mater. It typically involves arterial bleeding. In this patient, the middle meningeal artery may have been injured after a temporal bone fracture.

- A. Pia mater: This is the innermost meningeal layer, directly adherent to the brain. It is not linked to epidural hematomas.
- B. Arachnoid mater: This is the intermediate meningeal layer. It is not linked to epidural hemorrhages.
- D. Leptomeninges: This is a collective term (thin meninges) for the pia mater and arachnoid mater. Neither layer is linked to epidural hemorrhage.
- E. Choroid plexus: This is highly vascularized tissue that is distributed throughout the ventricular system of the brain. It is not linked to epidural hemorrhage.

16. A 72-year-old patient is brought to the physician because of a 3-month history of cognitive decline. He has a 5-year history of recurrent meningitis. MRI of the head shows enlarged ventricles, but no obstruction within the ventricular system or its major outflow foramina.

Which of the following structures is most likely dysfunctional in this patient?

- A. Pia mater
- B. Choroid plexus
- C. Arachnoid villi
- D. Denticulate ligaments
- E. Ligamentum flavum
- F. Cerebral aqueduct

Answer Key: C

The enlarged ventricles without obstruction indicate an imbalance between the production of cerebrospinal fluid (CSF) and its reabsorption. Arachnoid villi (or granulations) are the primary structures responsible for reabsorbing CSF from the subarachnoid space into the dural venous sinuses. Impaired function of these villi, often due to inflammation or fibrosis (e.g., from prior meningitis or subarachnoid hemorrhage), leads to communicating hydrocephalus.

A. Pia mater: This is the innermost meningeal layer, directly covering the brain and spinal cord. It does not play a role in CSF reabsorption from the subarachnoid space into the venous system.

B. Choroid plexus: This structure, located within the ventricles of the brain, is responsible for producing CSF. Impaired function could lead to decreased CSF production, not hydrocephalus. On the other hand, a choroidal tumor, in rare circumstances, can lead to increased production of CSF, but a tumor was not reported in this patient.

D. Denticulate ligaments: These are pial extensions that anchor the spinal cord to the dura mater, providing stability. They are not involved in CSF dynamics.

E. Ligamentum flavum: It is a ligament of the vertebral column, external to the dura, and completely unrelated to CSF production or reabsorption within the cranium.

F. Cerebral aqueduct: This structure connects the third and fourth ventricles. Obstruction here would cause non-communicating (obstructive) hydrocephalus, as CSF flow within the ventricular system would be blocked.

17. A 20-year-old man is brought to the emergency department because of a 1-hour history of a traumatic injury involving deceleration of the head. CT scan of the head shows brain damage associated with blood leaking from the anterior cerebral artery. Which of the following structures are most likely damaged in this patient?

- A. Brainstem and cerebellum
- B. Temporal lobe and Sylvian fissure
- C. Frontal and parietal lobes
- D. Occipital lobe and visual cortex
- E. Basal ganglia and thalamus

Answer Key: C

Portions of some major subarachnoid vessels form part of the circle of Willis and its branches. The anterior circulation primarily involves the internal carotid arteries (ICAs), anterior cerebral arteries (ACAs), and middle cerebral arteries (MCAs). The ACAs supply the medial aspects of the frontal and parietal lobes. The MCAs supply much of the lateral surface of the frontal, parietal, and temporal lobes, including the insular cortex and vicinity of the Sylvian fissure. A rupture in the ACA would most broadly affect the medial frontal and parietal regions.

A. Brainstem and cerebellum: These are primarily supplied by the posterior circulation (vertebral and basilar arteries).

B. Temporal lobe and Sylvian fissure: These structures are not tightly linked to the anterior cerebral arteries. The middle and posterior cerebral arteries would be more viable options.

D. Occipital lobe and visual cortex: Occipital regions are more closely linked to the posterior cerebral arteries. Note that much of the occipital lobes is devoted to visual processing.

E. Basal ganglia and thalamus: These deep prosencephalic structures are supplied by penetrating arteries, most commonly arising from the middle and posterior cerebral arteries.

18. During a neurosurgical procedure, a surgeon notes a small growth overlapping with a denticulate ligament. Which of the following descriptions best applies to the cellular and anatomical features of the affected structure in this patient?

- A. Nervous; lateral to spinal cord
- B. Meningeal; lateral to spinal cord
- C. Nervous; caudal to spinal cord
- D. Meningeal; caudal to spinal cord
- E. Nervous; rostral to spinal cord
- F. Meningeal; rostral to spinal cord

Answer Key: B

Denticulate ligaments are 21 pairs of flattened, triangular, tooth-like extensions of the pia mater that project laterally from the spinal cord. They pass through the arachnoid and attach to the inner surface of the dura mater, anchoring the spinal cord and providing it with lateral stability within the vertebral canal.

- A. The denticulate ligaments are non-nervous meningeal (pial) tissue.
- C. The denticulate ligaments are non-nervous meningeal (pial) tissue.
- D. While meningeal (pial), the denticulate ligaments are lateral to the spinal cord rather than caudal. The filum terminale is a pial ligament that is caudal to the spinal cord.
- E. The denticulate ligaments are non-nervous meningeal (pial) tissue.
- F. While meningeal (pial), the denticulate ligaments are lateral to the spinal cord rather than rostral. The brain is rostral to the spinal cord but not anchored by denticulate ligaments.

19. A 54-year-old man is brought to the emergency department because of a 20-minute history of unconsciousness. Physical examination shows ataxic breathing, consistent with damage to the caudal brainstem. CT scan of the head shows a subarachnoid hemorrhage. Which of the following arteries has most likely ruptured in this patient?

- A. Anterior cerebral
- B. Middle cerebral
- C. Internal carotid
- D. External carotid
- E. Posterior spinal

Answer Key: E

The brainstem typically receives blood via the vertebro-basilar arterial system. Vasospasm may have arisen secondary to the hemorrhage, further

promoting general bulbar dysfunction, with ataxic breathing and unconsciousness resulting. Even so, of the arteries listed, only the posterior spinal artery is a branch of the vertebral artery.

A. Anterior cerebral: The anterior cerebral artery does not supply blood to the brainstem. It arises from the circle of Willis and supplies the medial frontal and parietal lobes.

B. Middle cerebral: The middle cerebral artery does not supply blood to the brainstem. It supplies much of the lateral aspect of the cerebrum after branching from the circle of Willis.

C. Internal carotid: The internal carotid artery feeds the anterior and middle cerebral arteries at the level of the circle of Willis. It does not supply the brainstem.

D. External carotid: Much of the clinical relevance of the external carotid artery relates to the supply of dural vessels that may rupture with fracture of the skull, leading to epidural hemorrhage.

20. A neurosurgeon is planning to access a tumor located within the brain parenchyma. After incising the outer meninges, the surgeon encounters numerous small blood vessels closely adherent to the surface of the brain. Which of the following structures most likely covers the vessels under observation in this patient?

- A. Periosteal dura
- B. Meningeal dura
- C. Pia mater
- D. Arachnoid
- E. Falx cerebri

Answer Key: C

After incising the dura and arachnoid, the surgeon enters the subarachnoid space while avoiding the subarachnoid vessels. The pia mater is the innermost meningeal layer that directly adheres to the surface of the brain, following its contours precisely. The small blood vessels that supply the cerebral cortex (while receiving blood from the subarachnoid arteries) lie on the surface of the brain, intimately covered by the pia mater before penetrating the brain tissue.

- A. Periosteal dura: This layer is outermost and adheres to the inner surface of the skull.
- B. Meningeal dura: This layer rests between the periosteal dura and the arachnoid.
- D. Arachnoid: This structure serves as the outer boundary of the subarachnoid space.
- E. Falx cerebri: This structure rests between the cerebral hemispheres and is constructed from meningeal dura.

21. A 31-year-old woman is brought to the emergency department because of a 30-minute history of traumatic injuries stemming from a motor vehicle collision. A whiplash injury is diagnosed with involvement of the blood supply to the ventral myelon. Which of the following arteries is most likely affected in this patient?

- A. Vertebral
- B. Posterior spinal
- C. Anterior spinal
- D. Posterior inferior cerebellar
- E. Anterior inferior cerebellar

Answer Key: C

The myelon (spinal cord) is affected. The anterior spinal artery is a single artery that runs longitudinally along the anterior median fissure of the medulla and spinal cord. It supplies the anterior two-thirds of the spinal cord, including the anterior horns and the ventral and lateral and spinal white matter.

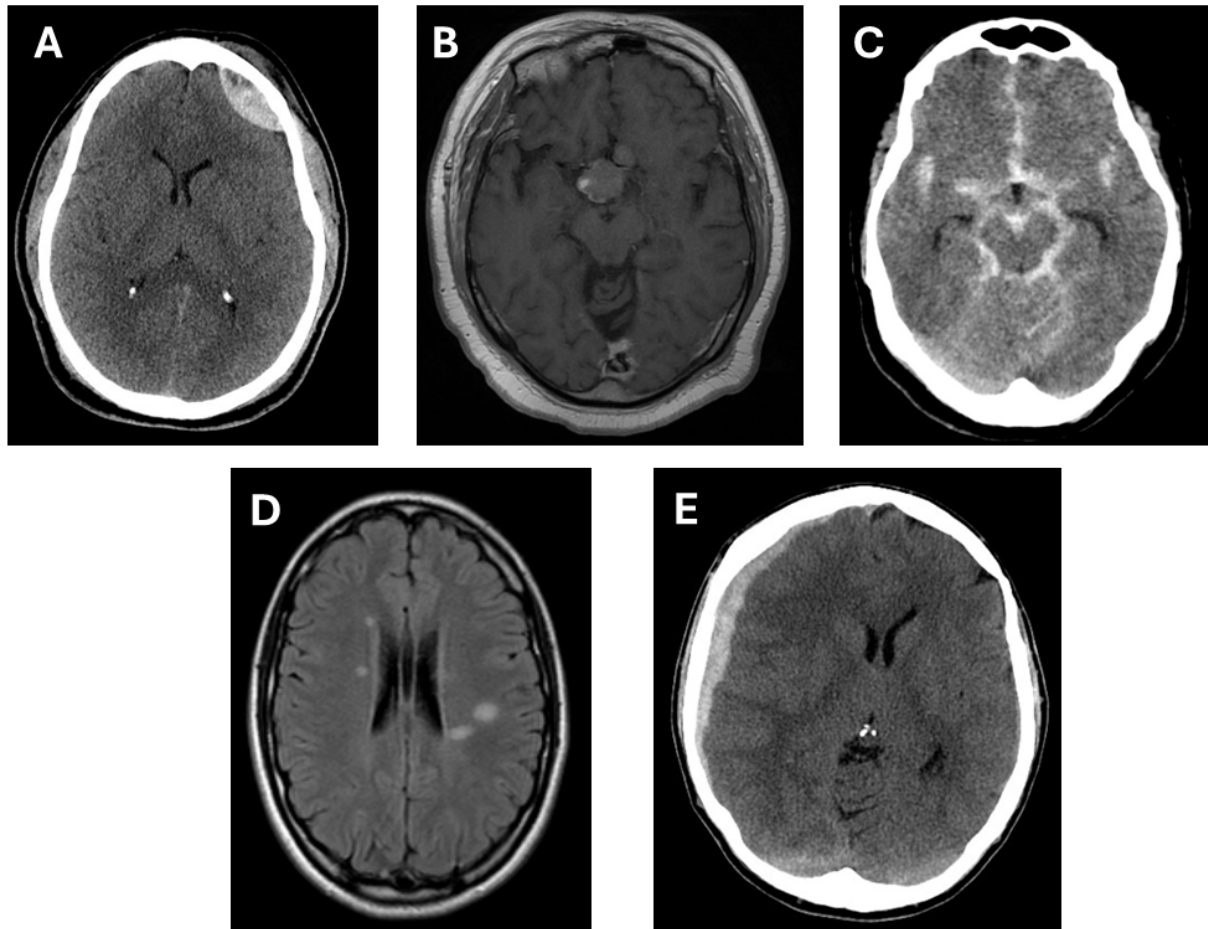
- A. Vertebral: The vertebral artery supplies the rostral extents of both the anterior and posterior spinal arteries.
- B. Posterior spinal: This artery supplies the dorsocaudal medulla and the entire length of the dorsal third of the spinal cord.
- D. Posterior inferior cerebellar: This artery supplies the rostrolateral medulla and portions of the cerebellum.
- E. Anterior inferior cerebellar: This artery supplies the caudolateral pons and portions of the cerebellum.

22. A 43-year-old man is brought to the emergency department by ambulance 2 hours after his husband found him lying at the foot of the stairs in a confused state. On arrival at the hospital, the patient is unconscious. Shortly after arrival, the patient briefly regains consciousness but loses consciousness again before the neurological examination can be completed. A swelling is evident in the right temporal region. CT scan of the head shows a right temporoparietal hyperdense, biconvex extra-axial lesion that does not cross the suture lines of the skull. Which of the following findings would most likely have been observed if neurological examination had been completed?
- A. Weakness on the right side of the body
 - B. Inability to comprehend spoken language
 - C. Non-fluent speech
 - D. Poor impulse control
 - E. Reduced somatosensation on the left

Answer Key: E

The history of recent blunt trauma, noticeable external swelling, and lucid interval are suggestive of an epidural hemorrhage. This is confirmed on neuroimaging, showing a characteristic biconvex accumulation that does not cross suture lines. Compression of the parietal lobe on the lateral convexity of the brain could involve the postcentral gyrus and would be predicted to lead to diminished somatosensation on the contralateral side of the body, option E (review NB Lecture 03, objective NSCI.0554). You should be able to predict which body part(s) would be most likely affected (review NB Lecture 02, objective NSCI.0021, primary somatosensory cortex). Language comprehension and production will likely lateralise to the left hemisphere, making options B and C doubtful. Impulse control localizes to the frontal lobe, while here there is visible trauma in the temporal region, along with a temporoparietal epidural hemorrhage. This makes option D improbable. Weakness on the right side of the body is unlikely since 1) the frontal lobe, responsible for voluntary movement, seems currently unaffected, and 2) damage at this level would produce contralateral deficits (review NB Lecture 03, objective NSCI.0554).

23. A 72-year-old man is brought to the physician by his wife because of a 5-week history of forgetfulness. His wife reports that her husband was previously meticulously organized and never forgot arrangements or belongings, but has recently had trouble remembering and finding things. The man recalls falling from a standing position approximately 6 weeks ago. He does not think he hit his head and did not require any medical attention. Neuroimaging shows a right-sided subdural hemorrhage. Which of the following images is most likely taken from this patient?



Images from <https://radiopaedia.org/>; full source details provided in the associated 'With Answers' document

Answer Key: E

Subdural hemorrhage has a characteristic crescent shape on neuroimaging. This type of lesion can be seen as a right-sided hyperdense region on the CT image in option E (see below for labelled images). Option A shows a biconvex epidural hemorrhage in the left frontal region. Option B shows a pituitary prolactinoma on axial T1-weighted MRI. Option C shows a subarachnoid hemorrhage on CT. Note the normal hypodense (dark) appearance of CSF in the lateral ventricles compared to the abnormal

hyperdense (bright) blood present in the subarachnoid space surrounding the midbrain and temporal lobes, and between the frontal lobes. Like the ventricles, the subarachnoid space should be filled with CSF and hence should appear dark on a CT image. Option D shows periventricular white matter lesions in FLAIR imaging in a case of multiple sclerosis (see NB DLA 01 Practical Clinical Neuroimaging III).

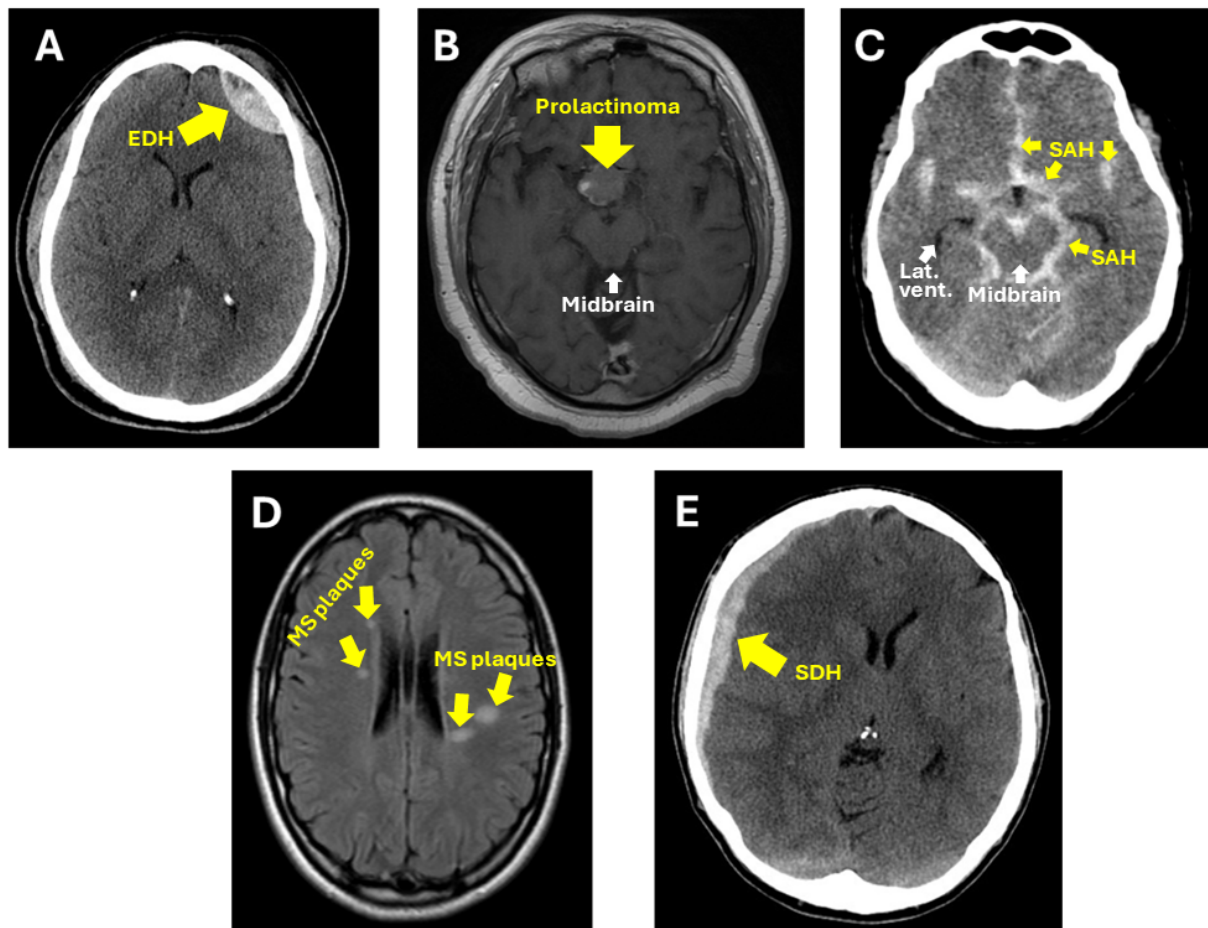


Image sources:

- A. Shif Y, Epidural haemorrhage. Case study, Radiopaedia.org (Accessed on 09 Sep 2025) <https://doi.org/10.53347/rID-26277>
- B. Thibodeau R, Pituitary prolactinoma. Case study, Radiopaedia.org (Accessed on 09 Sep 2025) <https://doi.org/10.53347/rID-166689>
- C. Puyó D, Subarachnoid haemorrhage. Case study, Radiopaedia.org (Accessed on 09 Sep 2025) <https://doi.org/10.53347/rID-22377>
- D. Abdrabou A, Multiple sclerosis with active lesions. Case study, Radiopaedia.org (Accessed on 09 Sep 2025) <https://doi.org/10.53347/rID-29069>
- E. Shif Y, Subdural haemorrhage. Case study, Radiopaedia.org (Accessed on 09 Sep 2025) <https://doi.org/10.53347/rID-26297>

NB 05 – Neurons and Glia

1. A 23-year-old man is brought to the emergency room because of a 4-hour history of unconsciousness. He dies 20 minutes later. Post-mortem studies reveal derangement of cells that normally contribute to the blood-brain barrier. Which of the following glial cells are most likely under study?
 - A. Astrocytes
 - B. Bipolar
 - C. Ependymal
 - D. Microglia
 - E. Oligodendrocytes
 - F. Schwann

Answer Key: A

One of the key functions of astrocytes is as a component of the blood-brain barrier. Bipolar cells are sensory cells, specifically for the special senses. Ependymal cells line the ventricles of the brain. Microglia are CNS macrophages whose key functions include phagocytosis and immune activation. Oligodendrocytes are the myelinating cells of the CNS, while Schwann cells are the myelinating cells of the PNS.

2. A neuroscience professor investigates the relationship between axonal transport and the onset of neurodegeneration. A defect in the anterograde axonal transport of which of the following is most likely implicated?
 - A. Cytoskeletal molecules
 - B. Lysosomes
 - C. Recycled vesicular membranes
 - D. Nerve growth factor

Answer Key: A

Cytoskeletal molecules undergo anterograde axonal transport. The other options all undergo retrograde axonal transport.

3. A 54-year-old woman comes to the physician because of numbness in both feet that has lasted for 2 days. She has a history of Type 2 diabetes mellitus. Neurological examination shows loss of touch and vibration sensation on both feet. Cells of which of the following morphotypes most likely affected in this patient?

- A. Pseudo-unipolar
- B. Unipolar
- C. Bipolar
- D. Multipolar

Answer Key: A

Pseudo-unipolar neurons are sensory neurons and are responsible for carrying information about pain, temperature, touch, vibration, and proprioception. Multipolar neurons include motor neurons, neurons in the autonomic ganglia, and interneurons. Bipolar neurons are responsible for the specialized senses of vision, smell, etc. Unipolar neurons are not present in humans.

4. In an experimental setting, axonal transport is assessed. Which of the following is most likely responsible for retrograde axonal transport?

- A. Actin
- B. Dynein
- C. Kinesin
- D. Neurofilament
- E. Ribosome

Answer Key: B

Dynein is a microtubule-associated protein responsible for retrograde axonal transport (towards the soma). Kinesin is also a microtubule-associated protein, but it is responsible for anterograde transport (away from the soma). Actin forms part of the cytoskeleton, while neurofilaments and ribosomes are involved in the maintenance of axonal diameter and protein synthesis, respectively.

5. A 68-year-old man comes to the physician because of a 4-week history of rapidly progressing weakness and muscle atrophy, initially in his hands. Neurologic examination shows weak myotatic reflexes. Clinical studies

show a disease process primarily affecting motor neurons, leading to widespread loss of the cellular structures responsible for transmitting signals over long distances to muscle fibers. Which of the following neuronal structures is most likely degenerating in this patient?

- A. Dendrite
- B. Soma
- C. Axon
- D. Axon terminal
- E. Nucleus

Answer Key: C

Progressive weakness, muscle atrophy, and hyporeflexia indicate disease of lower motor neurons. The description "transmitting signals over long distances to muscle fibers" directly refers to the function of the axon.

A. Dendrite: Dendrites primarily receive signals from other neurons, and they are not the main structure for long-distance signal transmission to muscles.

B. Soma: The soma (cell body) contains the nucleus and is crucial for cell survival and protein synthesis, but it is not the primary structure for long-distance transmission. While motor neuron cell bodies are affected in lower motor neuronal diseases such as ALS, the question focuses on long-distance transmission.

D. Axon terminal: Axon terminals of lower motor neurons form synapses and release neurotransmitters at the short distances associated with the neuromuscular junction.

E. Nucleus: The nucleus contains genetic material and controls cell function. While nuclear abnormalities can occur in neurodegenerative diseases, the nucleus does not transmit signals over long distances.

6. A 4-year-old boy is brought to the physician because of a 1-year history of developmental delays, seizures, and facial abnormalities. Genetic testing shows a disorder that results in the accumulation of undegraded macromolecules. Which of the following organelles is most likely dysfunctional in this patient?

- A. Mitochondrion
- B. Endoplasmic Reticulum

- C. Golgi Apparatus
- D. Lysosome
- E. Peroxisome

Answer Key: D

Lysosomal storage disorders are a group of inherited metabolic diseases characterized by the accumulation of various substrates within lysosomes due to enzyme deficiencies. Lysosomes are organelles containing digestive enzymes crucial for breaking down waste materials, cellular debris, and foreign substances. Their dysfunction leads to the accumulation described, causing cellular damage and clinical manifestations.

A. Mitochondrion: Mitochondria are primarily responsible for cellular respiration and ATP production. Dysfunctions lead to energy deficits, but not typically the accumulation of undegraded macromolecules.

B. Endoplasmic Reticulum: The endoplasmic reticulum (ER) is involved in protein synthesis (rough ER) and lipid synthesis/detoxification (smooth ER). While ER stress can occur in neurodegenerative diseases, its primary role is not waste breakdown and recycling.

C. Golgi Apparatus: The Golgi apparatus modifies, sorts, and packages proteins and lipids for secretion or delivery to other organelles. It is involved in cellular trafficking but is not the primary site of waste degradation.

E. Peroxisome: Peroxisomes are involved in fatty acid metabolism, detoxification, and reactive oxygen species breakdown. While peroxisomal disorders exist, the general accumulation of undegraded macromolecules from waste breakdown points to lysosomes.

7. A 72-year-old woman is brought to the physician because of a 6-month history of declining memory. She receives a diagnosis of Alzheimer's disease, which features impaired axonal transport, leading to dysfunctional neurotransmitter delivery to synaptic terminals. Which of the following structures is/are most likely deranged in this patient?

- A. Neurofilaments
- B. Microtubules
- C. Microfilaments
- D. Rough endoplasmic reticulum
- E. Cytoplasm

Answer Key: B

The question describes a patient with Alzheimer's disease exhibiting impaired axonal transport and dysfunctional neurotransmitter delivery. Axonal transport, which is the process of moving cellular components (like neurotransmitters, vesicles) along the axon, relies heavily on the microtubular cytoskeleton. Microtubules serve as the tracks along which motor proteins (like kinesins and dyneins) carry cargo. In Alzheimer's disease, pathological changes like tau hyperphosphorylation lead to the destabilization and aggregation of microtubules, directly impairing this vital transport system and contributing to synaptic dysfunction and neurodegeneration.

A. Neurofilaments: Neurofilaments are a type of intermediate filament abundant in axons, providing structural integrity and maintaining axonal caliber. While their disruption can occur in neurodegenerative diseases and affect axonal architecture, they are primarily structural elements and not the direct tracks for active long-distance axonal transport.

C. Microfilaments: Microfilaments are involved in cell shape, motility, and short-range transport, particularly at the periphery of the cell and within synaptic terminals (e.g., for vesicle recycling and actin dynamics at the synapse). However, they are not the primary long-distance transport tracks within the axon.

D. Rough endoplasmic reticulum: The rough endoplasmic reticulum (RER) is involved in the synthesis and folding of proteins, particularly those destined for

secretion or insertion into membranes. While neuronal function depends on proper protein synthesis by the RER, it is an organelle located primarily in the soma and dendrites and is not directly responsible for axonal transport itself. Dysregulation of protein synthesis can contribute to disease, but the question specifically points to the transport mechanism.

F. Cytoplasm: The cytoplasm is the entire contents within the cell membrane, excluding the nucleus. It is a broad term that includes organelles, the cytosol, and the cytoskeleton. While the derangement occurs within the cytoplasm, it is not a specific structure that is responsible for axonal transport; rather, microtubules (which are within the cytoplasm) are the specific structures at issue.

8. A neuroscientist is studying a novel cultured cortical interneuron that receives numerous inputs from various surrounding neurons and integrates these signals. This neuron exhibits a highly branched dendritic tree and a relatively short axon. What is the most likely morphological classification of this neuron?

- A. Bipolar
- B. Pseudounipolar
- C. Multipolar
- D. Pyramidal
- E. Unipolar

Answer Key: C

Multipolar neurons are the most common type of neuron in the central nervous system. They are characterized by an axon (that may collateralize) and profuse dendrites extending from the cell body. This morphology, particularly the highly branched dendritic tree, allows integration of vast inputs.

A. Bipolar: Bipolar neurons have two processes (axon and dendron) extending from opposite sides of the cell body. They are found in some sensory pathways (e.g., visual, auditory) but not typically in the cortex for integration.

B. Pseudounipolar: Cell bodies of pseudounipolar neurons emit a stalk that branches into two axonal segments (one central, one peripheral). They form peripheral sensory ganglia (e.g., dorsal root ganglia) and transmit sensory information from the periphery to the CNS.

D. Pyramidal: While pyramidal neurons are a type of large multipolar neuron found in the cortex (see other lectures for more details), they are characterized by a distinctive triangular soma and prominent apical dendrites, along with a relatively long axon that is thus capable of transmitting information great distances. The question describes a general interneuron and its overall branching pattern, making multipolar the broader and more direct morphological classification.

E. Unipolar: Unipolar neurons have only one process extending from the cell body. These are primarily found in invertebrates.

9. A 28-year-old woman comes to the physician because of a 3-year history of vision loss, numbness, and weakness. Her neurological symptoms are

attributed to demyelination in the central nervous system. Which of the following processes is most likely impaired in this patient?

- A. Propagation of action potentials
- B. Synaptic transmission
- C. Anterograde axonal transport
- D. Diffusion
- E. Retrograde axonal transport

Answer Key: A

Multiple sclerosis is an autoimmune disease characterized by demyelination in the CNS. Myelin sheaths are lipid-rich coverings that insulate axons and facilitate the propagation of action potentials. Demyelination disrupts this process, significantly slowing or blocking axonal impulse transmission.

B. Synaptic transmission: Mechanisms directly linked to synaptic transmission are not dependent on myelination. However, the failure of action potentials to propagate normally down axons indirectly affects synaptic transmission.

C. Anterograde axonal transport: Anterograde axonal transport involves microtubules and a specialized motor protein (i.e., kinesin).

D. Diffusion: Diffusion refers to the random movement of dissolved or suspended particles linked to molecular collisions.

E. Retrograde axonal transport: Retrograde axonal transport involves microtubules and a specialized motor protein (i.e., dynein).

10. A 33-year-old woman comes to the physician because of a 2-hour history of a deep penetrating wound to her left arm. Neurologic examination shows ulnar nerve injury. It is expected that, distal to the injury, the axon and myelin will degenerate. Which of the following central cellular reactions is most likely to arise early in this patient?

- A. Wallerian degeneration
- B. Apoptosis
- C. Transneuronal degeneration
- D. Chromatolysis
- E. Microglial activation

Answer Key: D

Axonal transection initiates a series of cellular reactions. Chromatolysis is the characteristic change observed in the neuronal cell body (soma) following axonal injury. It involves swelling of the soma, dispersion of the rough endoplasmic reticulum (Nissl bodies), and eccentric displacement of the nucleus. In this case, the central cell bodies of motor axons forming part of the ulnar nerve would be involved. The process of Wallerian degeneration is described in the question, but this process would not be central, noting the damage to the ulnar nerve in the periphery.

A. Wallerian degeneration: Wallerian degeneration primarily involves distal axons and myelin. The question asks about the cellular reaction in the central nervous system.

B. Apoptosis: Apoptosis is programmed cell death. While severe or prolonged injury can lead to apoptosis, chromatolysis is a specific morphological change indicating a metabolic response to injury, often preceding attempts at repair.

C. Transneuronal degeneration: This refers to the slow-onset degeneration of neurons that are synaptically connected to the injured neuron (either upstream or downstream), not the primary reaction of the injured neuron's own cell body.

E. Microglial activation: Microglial activation is part of an inflammatory response that occurs in both proximal and distal segments of central cells and axons. This could eventually arise centrally following peripheral ulnar injury in reaction to slower-onset secondary degeneration of central neurons.

11. A 40-year-old man develops progressive ascending muscle weakness and areflexia following a viral infection. Lumbar puncture reveals elevated CSF protein with normal cell count. Electrodiagnostic studies show demyelination of peripheral nerves. Which of the following cell types is most likely affected in this patient?

- A. Oligodendrocytes
- B. Astrocytes
- C. Schwann cells
- D. Microglia
- E. Ependymal cells

Answer Key: C

The clinical presentation (ascending weakness, areflexia, post-infectious onset, elevated CSF protein with normal cell count, peripheral nerve demyelination) suggests Guillain-Barré Syndrome (GBS). GBS is an autoimmune polyneuropathy where the immune system mistakenly attacks Schwann cells in the peripheral nervous system (PNS), leading to demyelination and impaired nerve conduction.

A. Oligodendrocytes: Oligodendrocytes produce myelin in the central nervous system (CNS). Demyelination of oligodendrocytes is characteristic of diseases like multiple sclerosis, not GBS, which affects peripheral nerves.

B. Astrocytes: Astrocytes are CNS glial cells involved in support, the blood-brain barrier, and extracellular regulation. They are not directly involved in PNS myelin production nor the primary pathology of GBS.

D. Microglia: While immune activation is involved in GBS, microglia are centrally and not peripherally active.

E. Ependymal cells: Ependymal cells are associated with the central ventricular system. They are not related to PNS myelin nor to GBS pathology.

12. A 60-year-old man comes to the physician because of a 6-month history of headaches and nausea with recurrent seizures. MRI of the head shows a tumor of the brain. Laboratory studies show that the tumor is primarily composed of cells forming extensive and abnormal myelin-like structures. Which of the following populations of glia cells are most likely implicated in the tumor in this patient?

- A. Astrocytes
- B. Oligodendrocytes
- C. Ependymal cells
- D. Schwann cells
- E. Microglia

Answer Key: B

Oligodendrocytes are the glial cells responsible for producing and maintaining the myelin sheath around central axons. A tumor composed of cells forming abnormal myelin-like structures within the brain strongly suggests an origin from oligodendrocytes, consistent with an oligodendroglioma, a type of primary brain tumor.

A. Astrocytes: Astrocytes are involved in support/regulatory functions and the blood-brain barrier. While astrocytomas are relatively common, they typically do not form myelin-like structures.

C. Ependymal cells: Ependymal cells line the brain ventricles and produce CSF. Tumors arising from these cells (i.e., ependymomas) are therefore not characterized by myelin-like formations.

D. Schwann cells: Schwann cells produce myelin, but they are located in the periphery. The tumor is described as a brain lesion, placing it in the CNS. Note, however, that schwannomas can grow such that they eventually compress central tissue.

E. Microglia: Microglia are immune cells of the CNS. Tumors involving microglia are rare and would not be characterized by myelin-like formations.

NB 06 – Transport within the CNS

1. A 4-year-old boy is brought to the emergency department because of a 1-day history of fever, headache, and neck stiffness. A diagnosis of bacterial meningitis is made following a lumbar puncture for CSF analysis. Which of the following is the most appropriate site for a lumbar puncture in this patient?

- A. L1/L2
- B. L2/L3
- C. L3/L4
- D. L4/L5

Answer Key: D

The spinal cord normally ends at the L3 vertebral level in children.

Therefore, a lumbar puncture is ideally done at the L4/L5 vertebral level. In adults, the procedure is done at a higher level (L3/L4) since the spinal cord ends at L1/L2.

2. A 67-year-old woman is brought to the physician because of memory problems that started about 3 months ago. Following assessment, a diagnosis of Alzheimer's disease is made. Imaging shows

hydrocephalus. Which of the following is most likely present in this patient?

- A. Communicating hydrocephalus
- B. Non-communicating hydrocephalus
- C. Obstructive hydrocephalus
- D. Hydrocephalus ex vacuo
- E. Normal pressure hydrocephalus

Answer Key: D

This patient most likely has hydrocephalus ex vacuo. Due to the presence of a neurodegenerative disease in this patient, the ventricles tend to enlarge as brain tissue shrinks.

3. A 1-week-old child is brought to the physician because of a rapid increase in the size of the head since birth. Examination and imaging studies are consistent with hydrocephalus due to aqueductal stenosis. Which of the following is most likely present in this patient?

- A. Communicating hydrocephalus
- B. Non-communicating hydrocephalus
- C. Hydrocephalus ex vacuo
- D. Normal pressure hydrocephalus

Answer Key: B

This patient most likely has non-communicating (obstructive) hydrocephalus due to the obstruction to CSF flow from the cerebral aqueduct stenosis.

4. A 74-year-old man is brought to the emergency department because of a 30-minute history of severe nausea and vomiting. Examination and investigation findings are consistent with the involvement of a circumventricular organ. Which of the following is the most likely involved in this patient's presentation?

- A. Amygdala
- B. Area postrema

- C. Basal ganglia
- D. Cerebellum
- E. Hippocampus

Answer Key: B

The area postrema is a circumventricular organ located in the medulla. It is known as the vomiting center. The other options are not circumventricular organs. The amygdala plays a role in emotion, fear, and memory. The hippocampus is involved in memory and learning. The basal ganglia and cerebellum are responsible for the modulation of voluntary motor activity.

5. A 1-month-old child is brought to the physician because of an enlarged head, poor feeding, and irritability. A diagnosis of hydrocephalus is made following imaging studies, which show enlarged lateral and third ventricles, with a normal-sized fourth ventricle. Which of the following is most likely responsible for the findings in this patient?
- A. Brain tissue atrophy
 - B. Impaired CSF absorption
 - C. Obstruction of the cerebral aqueduct
 - D. Obstruction of the intraventricular foramen

Answer Key: C

Imaging studies show the presence of an obstruction at the cerebral aqueduct (enlarged lateral and third ventricles, with a normal sized fourth ventricle). This is an obstructive hydrocephalus. Brain tissue atrophy can be seen in hydrocephalus ex vacuo, while impaired CSF absorption can cause communicating hydrocephalus.

6. A 6-month-old child is brought to the clinic with progressive head enlargement and irritability. Examination reveals bulging fontanelles. MRI shows dilation of all ventricles without evidence of obstruction. The physician suspects impaired cerebrospinal fluid absorption. Which structure is most likely dysfunctional?
- A. Choroid plexus of the lateral ventricles

- B. Foramina of Monro (interventricular foramina)
- C. Cerebral aqueduct (of Sylvius)
- D. Arachnoid villi in the superior sagittal sinus
- E. Subarachnoid cisterns

Answer Key: D

Arachnoid villi (and granulations) drain cerebrospinal fluid (CSF) into the dural venous sinuses. Dysfunction of these structures can impair CSF absorption, leading to communicating hydrocephalus characterized by dilation of all ventricles.

7. A 4-year-old child is brought to the emergency department with signs of increased intracranial pressure: headache, vomiting, and papilledema. CT imaging reveals overproduction of CSF due to a tumor. Which site is most likely the origin of this tumor?

- A. Ependymal lining of the cerebral aqueduct
- B. Choroid plexus of the lateral ventricles
- C. Arachnoid villi in the superior sagittal sinus
- D. Astrocytes in the subdural space

Answer Key: B

The choroid plexus is a specialized epithelium that produces CSF through plasma filtration and active transport. A choroid plexus papilloma can cause CSF overproduction, resulting in hydrocephalus.

NB 07 – Stimulated Excitable Cells

1. A 23-year-old woman comes to the physician because of a 3-week history of tingling affecting her left leg. Imaging studies show a central inflammatory demyelinating lesion is found in an ascending somatosensory pathway. She receives a diagnosis of multiple sclerosis. What are the functional effects of the lesion on the electrical properties of the affected axons?
- A. Decreased membrane resistance and decreased length constant
 - B. Decreased membrane resistance and increased length constant
 - C. Increased membrane resistance and decreased length constant
 - D. Increased membrane resistance and increased length constant

Answer Key: A

The demyelinating lesion compromises the electrical insulation of axons, thus functionally reducing the electrical resistance of the membrane.

Because the membrane is now leakier, the length constant of the affected axons is reduced.

2. A 64-year-old woman comes to the physician because of a 1-week history of weakness and numbness in her legs. Nerve conduction studies show reduced conduction velocity in her peripheral nerves. Which of the following is most likely responsible for the nerve conduction study findings in this patient?

- A. Increased axonal diameter
- B. Increased myelination
- C. Decreased myelination
- D. Decreased intracellular resistance
- E. Increased cross-sectional area

Answer Key: C

Myelination improves conduction speed. Therefore, decreased myelination slows conduction speed. Increased axonal diameter increases conduction speed as well. This results in a larger cross-sectional area > low intracellular resistance.

3. A 25-year-old man comes to the physician because of dizziness and palpitations that have lasted for 4 days. Examination and ECG findings are consistent with Long QT syndrome. A defect in the S4 transmembrane domain is implicated. Which of the following is a feature of the affected transmembrane domain?

- A. Negatively charged
- B. Non-helical structure
- C. Confers voltage sensitivity
- D. Induces channel closing

E. Inhibits sodium flow

Answer Key: C

The S4 transmembrane domain confers voltage sensitivity. It is also a positively charged helical structure that enhances sodium flow by inducing channel opening.

4. A 40-year-old man comes to the physician because of a 5-day history of knee pain and trouble straightening his left leg. During surgery for the removal of a ruptured medial meniscus, a drug is delivered into the lumbar subarachnoid space. Anesthesia and paralysis of the legs arise. Which of the following structures is the most likely site of action of the drug?

- A. Acetylcholine receptors
- B. GABA receptors
- C. Glutamate receptors
- D. Glycine receptors
- E. Voltage-gated sodium channels
- F. Voltage-gated potassium channels
- G. Voltage-gated calcium channels
- H. Voltage-gated chloride channels

Answer Key: E

Local anesthetics block voltage-gated sodium channels on axons. This does not affect receptor or post-synaptic potentials. However, action potentials will be prevented.

5. A 23-year-old man is brought to the emergency department because of a cut he sustained on his forearm while at work. Lidocaine is administered before the wound is sutured. Which of the following best explains the mechanism of action of the administered local anesthetic?

- A. Calcium channel block
- B. Increase in action potential
- C. Potassium channel block

- D. Sodium channel block
- E. Stimulation of calcium channels

Answer Key: D

Lidocaine is a commonly used local anesthetic that blocks voltage-gated sodium channels, resulting in the suppression of action potential.

6. A 53-year-old man comes to the physician because of a 36-hr history of headaches. Neurologic examination shows normal somatosensory function. When the physician applies pressure to the skin of the patient, somatosensory endings depolarize in a graded fashion. Which of the following electrical potentials is most likely under consideration in this patient?

- A. Action
- B. Excitatory postsynaptic
- C. Inhibitory postsynaptic
- D. Receptor
- E. Resting membrane

Answer Key: D

A receptor potential is a type of graded potential that is generated in sensory receptor cells or neurons in response to a specific stimulus (like touch). It reflects a local non-propagating electrical flux, the amplitude of which varies with the intensity of the stimulus and which only spreads decrementally.

- A. Action: An action potential is an all-or-none self-propagating electrical signal, typically initiated at the axon hillock.
- B. Excitatory postsynaptic: An excitatory postsynaptic potential is a depolarizing graded potential, but it typically arises at synapses and is thus distinct from a receptor potential.
- C. Inhibitory postsynaptic: An inhibitory postsynaptic potential is a graded potential, but it is typically associated with induction at synapses and is generally repolarizing or hyperpolarizing in nature. It is thus distinct from a receptor potential.
- E. Resting membrane: Resting membrane potential is the tonic level of polarization of excitable cells. It is determined by basal differential

distributions of ions (chiefly sodium and potassium) across semipermeable membranes.

7. A 41-year-old woman is brought to the emergency department because of a 30-minute history of envenomation. Based on the description of the envenomating organism, it is assumed that a toxin is blocking acetylcholine receptors at the neuromuscular junction. Which of the following events is most likely directly impacted by the toxin in this patient?
- A. Activation of voltage-gated sodium channels
 - B. Activation of voltage-gated potassium channels
 - C. Activation of ligand-gated sodium channels
 - D. Activation of ligand-gated potassium channels
 - E. Inactivation of voltage-gated sodium channels
 - F. Inactivation of voltage-gated potassium channels

Answer Key: C

At the neuromuscular junction, acetylcholine is released by lower motor neurons, with end-plate potentials then arising consequent to the opening of the associated ligand-gated (acetylcholine-gated) sodium channels.

Blockage of the nicotinic acetylcholine receptors found at the neuromuscular junction thus prevents the influx of sodium normally triggered by the binding of acetylcholine to this receptor type.

- A. Activation of voltage-gated sodium channels: Activation of voltage-gated sodium channels can arise subsequent to suprathreshold graded depolarization.
- B. Activation of voltage-gated potassium channels: Activation of voltage-gated potassium channels arises secondary to high levels of depolarization associated with the onset of action potentials.
- D. Activation of ligand-gated potassium channels: Activation of ligand-gated potassium channels is not a phenomenon that is observed in the vertebrate neuromuscular junction, as the vertebrate somatic muscles only express excitatory endplate potentials and not inhibitory endplate potentials.
- E. Inactivation of voltage-gated sodium channels: Voltage-gated sodium channels inactivate momentarily following activation during action potentials.

F. Inactivation of voltage-gated potassium channels: Voltage-gated potassium channels inactivate later during action potentials to oppose repolarization or hyperpolarization.

8. A research study investigates a genetic mutation that causes an abnormally leaky phospholipid bilayer. Such a defect would most directly impact the passive spread of electrical signals, leading to their rapid dissipation from the site of origin. Which of the following changes to cellular parameters is most likely in this context?

- A. Increased membrane resistance
- B. Decreased membrane resistance
- C. Increased intracellular resistance
- D. Decreased intracellular resistance

Answer Key: B

The length constant (λ) relates to the distance along which an electrical signal (like a graded potential) can spread passively along a neuronal process (dendrite or axon) before decaying to 37% of its original amplitude. Relevant to the length constant is the square root of the ratio of membrane resistance over intracellular resistance. The scenario involves a leaky cellular membrane, implying that the electrical resistance of the membrane is pathologically decreased, leading to a decreased length constant.

- A. Increased membrane resistance: Increased membrane resistance would culminate in an increased rather than in a decreased length constant.
- C. Increased intracellular resistance: Increased intracellular resistance would decrease the length constant but would not relate to the integrity of the cellular membrane.
- D. Decreased intracellular resistance: Decreased intracellular resistance would increase the length constant, but also would not relate to the integrity of the cellular membrane.

9. In an experimental setting, researchers modify neuronal properties such that larger excitatory postsynaptic potentials are required to provoke

action potentials. Which of the following sites is most likely being manipulated by the researchers?

- A. Axonal terminal
- B. Postsynaptic receptor
- C. Myelin
- D. Axon hillock
- E. Synaptic vesicle

Answer Key: D

The axon hillock is the spike-initiating zone in neurons because it has the highest density of voltage-gated sodium channels (giving it the lowest threshold for excitation). It also houses the population of voltage-gated sodium channels that sit nearest to the source of excitatory postsynaptic potentials. An increasing threshold for the initiation of action potentials could therefore relate to the responsiveness of voltage-gated sodium channels situated at the axon hillock.

A. Axonal terminal: The axonal terminal releases transmitter in reaction to depolarization. While changes in function of the axonal terminal could affect transmitter release and hence the magnitude of postsynaptic potentials, they would not be expected to impact function at the axon hillock of a postsynaptic cell.

B. Postsynaptic receptor: Altered function of the postsynaptic receptor would be expected to impact the magnitudes of postsynaptic potentials but not impact the threshold for action potentials.

C. Myelin: While myelin is an important regulator of propagation of action potentials, it is not clear that altered function of myelin should impact the threshold for the firing of action potentials.

E. Synaptic vesicle: Altered function of synaptic vesicles would be expected to impact the amount of transmitter released from presynaptic terminals. However, postsynaptic reactions to the release of transmitter ought not to be impacted in a direct fashion.

10. A 6-year-old boy is brought to the physician because of a 2-month history of muscle weakness and periodic paralysis. Genetic testing reveals a mutation that causes voltage-gated sodium channels to inactivate slightly more slowly than normal. Which of the following electrical phenomena is most likely prolonged in this patient?

- A. Repolarization
- B. Hyperpolarization
- C. Depolarization
- D. Resting state

Answer Key: C

The rapid depolarization phase of the action potential is primarily driven by the fast influx of sodium ions through voltage-gated sodium channels. If these channels inactivate more slowly, the influx of sodium would continue for a longer duration, thus directly prolonging the depolarization phase while perhaps only secondarily prolonging repolarization (depending on the efficacy of voltage-gated potassium channels).

A. Repolarization: Repolarization is primarily driven by the efflux of potassium ions. Slower sodium channel inactivation would only impact repolarization indirectly, and it seems possible that that potassium channels could overcome prolongation of repolarization secondary to reduced activation of voltage-gated sodium channels.

B. Hyperpolarization: Hyperpolarization (after-hyperpolarization) occurs after repolarization, due to sustained potassium efflux. Slower sodium inactivation would not directly prolong this phase.

D. Resting state: The resting state is the period before an action potential. The mutation affects channel behavior during depolarization, not the resting state.

11. A 22-year-old woman comes to the physician because of a 48-hour history of progressive weakness. The underlying pathology relates to disrupted segments of axonal insulation. Which of the following cellular components is most likely under attack in this patient?

- A. Voltage-gated potassium channels
- B. Ligand-gated channels
- C. Nodes of Ranvier
- D. Voltage-gated sodium channels
- E. Myelin sheaths

Answer Key: E

In rapidly propagating axons, specialized glial cells form myelin. Between the myelinated axonal segments, nodal regions express high concentrations of voltage-gated sodium channels. Disruption of the axonal insulation electrically disconnects nodal regions from each other, thus slowing or blocking saltatory conduction (and thereby leading to weakness in this patient).

A. Voltage-gated potassium channels: While voltage-gated potassium channels are present near nodes and contribute to repolarization, they do not participate in segmental insulation of axons.

B. Ligand-gated channels: Ligand-gated channels are typically found at synapses and respond to neurotransmitters and are thus not involved in segmental myelination.

C. Nodes of Ranvier: Nodes of Ranvier are the anatomical locations where the action potential is regenerated (as part of saltatory conduction). The nodes lie between myelinated axonal segments.

D. Voltage-gated sodium channels: Voltage-gated sodium channels are situated in the uninsulated (nodal) regions in myelinated axons.

12. In an experimental setting, the propagation of action potentials along axons with different properties is compared. The first axon is large and myelinated, whereas the second axon is small and unmyelinated. Which of the following descriptions best applies to the comparison of the propagating potentials reported in this study?

A. Graded vs. all-or-none

B. Saltatory vs. continuous

C. Depolarization vs. repolarization

D. Orthodromic vs. antidromic

E. Inhibitory vs. excitatory

Answer Key: B

In myelinated axons, action potentials propagate via saltatory conduction (jumping) sequentially between nodes of Ranvier. In unmyelinated axons, action potentials propagate via slower continuous conduction, regenerating at every point of the axonal membrane.

A. Graded vs. all-or-none: This differentiates between subthreshold local potentials and action potentials, not the propagation method along

different axon types. Both types of axons (myelinated and unmyelinated) conduct all-or-none action potentials.

C. Depolarization vs. repolarization: These are phases of an action potential itself, not different modes of propagation along different axonal types.

D. Orthodromic vs. antidromic: This describes the direction of propagation (normal physiological direction vs. reverse direction) and not the mechanism of propagation. Under physiological conditions, orthodromic conduction is expected. However, under certain abnormal circumstances (e.g., during clinical or accidental electrical stimulation of axons), antidromic conduction/propagation of action potentials can also occur.

E. Inhibitory vs. excitatory: This describes the effect of a neuron on its postsynaptic target, not the mechanism of action potential propagation along its own axon.

13. A 2-year-old boy is brought to the physician because of a 2-month history of reduced mobility and muscle stiffness. He receives a diagnosis of a rare genetic disorder characterized by hyperexcitable lower motor neurons. Electrophysiological studies show that these neurons fire action potentials in response to very weak stimuli that would normally be subthreshold. Which of the following phenomena is most likely in the lower motor neurons of this patient?

- A. Tonic depolarization
- B. Increased action potential amplitude
- C. Elevated threshold for excitation
- D. Increased conduction velocity
- E. Decreased membrane resistance

Answer Key: A

Neuronal hyperexcitability, particularly firing in response to subthreshold stimuli, indicates a relative lowering of the threshold for excitation. The threshold is the critical membrane potential that must be reached to trigger an all-or-none action potential. However, a lowered threshold for the firing of action potentials is not listed among the possible answer choices. However, were the motor neurons to be persistently depolarized (perhaps due to compromised inhibitory input), then lesser magnitudes of stimulation would be required to provoke action potentials.

- B. Increased action potential amplitude: The action potential is an all-or-none event, meaning its amplitude is largely fixed during propagation, regardless of stimulus strength above threshold.
- C. Elevated threshold for excitation: The stem of the question explicitly indicates that the threshold for excitation of the lower motor neurons are is relatively lowered and not raised.
- D. Increased conduction velocity: Conduction velocity refers to the speed of propagation along the axon after the action potential is generated and not to the ease of its initiation.
- E. Decreased membrane resistance: While changes in membrane resistance can influence excitability, decreased membrane resistance would be associated with a reduced length constant, perhaps making it less likely that graded dendritic or somatic depolarization could trigger action potentials.

14. A 30-year-old woman comes to the physician because of a 2-week history of weakness, particularly affecting her lower limbs. Laboratory studies show diffuse peripheral axonal degeneration with a corresponding loss of axonal diameter. Which of the following changes is most likely evident in this patient?

- A. Increased intracellular resistance
- B. Decreased intracellular resistance
- C. Increased membrane resistance
- D. Decreased membrane resistance

Answer Key: A

Axonal diameter is inversely related to intracellular resistance. Therefore, axons with lesser diameters (and hence lesser cross-sectional areas) will express higher levels of intracellular resistance.

- B. Decreased intracellular resistance: Axonal diameter is inversely related to intracellular resistance. Therefore, axons with lesser diameters (and hence lesser cross-sectional areas) will express higher levels of intracellular resistance and not lower levels of intracellular resistance.
- C. Increased membrane resistance: Resistance of the membrane ought to be unaffected by axonal diameter.
- D. Decreased membrane resistance: Resistance of the membrane ought to be unaffected by axonal diameter.

15. In an experimental setting, a novel neurotoxin binds to the inactivation gate of voltage-gated sodium channels in cultured neurons. Which of the following changes to the dynamics of action potentials is most likely in this experimental preparation?
- A. Enhanced repolarization
 - B. Enhanced hyperpolarization
 - C. Sustained depolarization
 - D. Accelerated depolarization
 - E. Reduced action potential amplitude

Answer Key: C

The inactivation gate of voltage-gated sodium channels is responsible for closing the channel shortly after it opens during depolarization, which is crucial for initiating repolarization and the refractory period. If this gate cannot close, sodium ions would continue to flow into the cell, leading to a prolonged or sustained depolarization of the membrane, unless sufficiently overpowered by potassium channels.

- A. Enhanced repolarization: Slower inactivation of sodium channels opposes the repolarizing influence of potassium channels during action potentials.
- B. Enhanced hyperpolarization: Hyperpolarization is typically caused by potassium efflux or chloride influx. Preventing sodium channel inactivation would keep the cell depolarized, not hyperpolarized.
- D. Accelerated depolarization: The toxin affects channel inactivation after depolarization has occurred and not the rate at which depolarization occurs once the threshold is reached.
- E. Reduced action potential amplitude: Throughout the preterminal portions of the axon, the amplitude of an action potential is stereotyped in size. Preventing inactivation of voltage-gated sodium channels would promote rather than oppose depolarization.

16. In an experimental setting, radiolabeled antibodies against voltage-gated sodium channels are applied to cultured neurons. Which of the following structures are most likely targeted?

- A. Axon terminals

- B. Nodes of Ranvier
- C. Myelin segments
- D. Somata
- E. Dendrites

Answer Key: B

In myelinated peripheral nerves, voltage-gated sodium channels are highly concentrated at the nodes of Ranvier. These are the sites where the action potential is regenerated during saltatory conduction. If these channels are targeted by antibodies (as in an autoimmune disorder), the ability of action potentials to jump (saltate) and thus propagate efficiently from node to node would be severely impaired.

- A. Axon terminals: Axon terminals contain voltage-gated calcium channels primarily for neurotransmitter release. While sodium channels are present, their highest density for propagation in myelinated axons is at the nodes.
- C. Myelin segments: Myelin segments are insulating and express very few ion channels. Their role is to increase resistance and decrease capacitance between nodes and not to generate action potentials.
- D. Somata: Cell bodies express few voltage-gated sodium channels.
- E. Dendrites: Dendrites express few voltage-gated sodium channels.

NB 08 – Synaptic Transmission

1. A neuroscience professor conducting a study on synapses investigates the features of electrical synapses. Which of the following is the most likely feature of the synapse of interest?
 - A. Gap junctions
 - B. Slow intercellular communication
 - C. Indirect intercellular movement of ions
 - D. Communication between cells is always unidirectional

Answer Key: A

Electrical synapses are mediated by gap junctions. They are also characterized by rapid intercellular communication and pass ions directly between cells. Communication can be bidirectional.

2. In an experimental setting, the quantal release of neurotransmitters is studied. Which of the following best describes the quantal content?
- A. Content of synaptic vesicles
 - B. Transmitter released from a single vesicle
 - C. Number of vesicles per impulse
 - D. Smallest postsynaptic response
 - E. Number of impulses for a single vesicle

Answer Key: C

The quantal content is the number of vesicles (quanta) per impulse. On the other hand, the quantal size is the smallest post-synaptic response and relates to the transmitter released from a single vesicle.

3. A 37-year-old man comes to the emergency department because of a 4-hour history of traumatic injury to the foot. Emergency surgery is prescribed. An anesthetist administers an acetylcholine receptor blocker to induce paralysis. Which of the following changes in synaptic transmission is most likely?
- A. Reduced quantal content
 - B. Reduced quantal size
 - C. Increased quantal content
 - D. Increased quantal size

Answer Key: B

The receptor blocker does not alter the number of vesicles undergoing exocytosis per impulse. However, the post-synaptic response to the release of a quantum would be diminished, as there would be fewer available binding sites for the molecules of transmitter released from a given vesicle.

4. A neurotoxicologist researching the effects of various toxins on neurotransmission isolates a toxin that inhibits nicotinic acetylcholine receptors in the post-synaptic membrane. Which of the following is the most likely toxin studied?

- A. Botulinum toxin
- B. W-conotoxin
- C. Tetanus toxin
- D. Alpha bungarotoxin
- E. Beta bungarotoxin

Answer key: D

Alpha bungarotoxin, a component of snake venom, binds to the nicotinic acetylcholine receptors on the postsynaptic membrane, thereby inhibiting the binding of acetylcholine and resulting in paralysis.

5. A 58-year-old woman with a 5-month history of cervical dystonia (neck muscle spasms) comes to the physician for a follow-up. She is administered Myobloc (botulinum toxin type B). Which of the following is the most likely effect of this agent?
- A. Blockade of acetylcholine receptors
 - B. Blockade of calcium channels
 - C. Blockade of sodium channels
 - D. Inhibition of acetylcholine release
 - E. Inhibition of acetylcholinesterase

Answer Key: D

Botulinum toxin is used for the treatment of a variety of conditions, including cervical dystonia. It results in the breakdown of synaptobrevin (which plays a role in the fusion of synaptic vesicles with the presynaptic membrane and the subsequent neurotransmitter release). The toxin, therefore, prevents the release of acetylcholine.

6. A 45-year-old man comes to the physician because of a 3-month history of muscle weakness and difficulty climbing stairs. Electromyography shows an increasing response to repetitive nerve stimulation. The patient is diagnosed with Lambert-Eaton myasthenic syndrome, which is caused by an immune attack on voltage-gated calcium channels. Which of the following phenomena best explains the weakness in this patient?

- A. Decreased quantal size
- B. Slower presynaptic repolarization
- C. Increased probability of vesicle fusion
- D. Reduced numbers of synaptic vesicles released per action potential
- E. Prolonged postsynaptic depolarization

Answer Key: D

The arrival of an action potential at the presynaptic terminal opens voltage-gated calcium channels. The subsequent influx of calcium ions triggers the fusion of synaptic vesicles with the membrane. A decrease in the number of functional calcium channels directly reduces the calcium influx and, therefore, the quantal content (number of vesicles released per impulse).

A. Decreased quantal size: Quantal size refers to the smallest measurable postsynaptic reaction to transmitter released from the presynaptic terminal. This is not directly affected by the number of open calcium channels, which instead impacts the number of vesicles released.

B. Slower presynaptic repolarization: Repolarization of the presynaptic terminal is primarily mediated by voltage-gated potassium channels. The slowing of presynaptic repolarization should increase the number of open calcium channels, thereby reversing weakness associated with Lambert-Eaton syndrome.

C. Increased probability of vesicle fusion: Increased probability of vesicle fusion should augment transmitter release and diminish weakness associated with Lambert-Eaton syndrome.

E. Prolonged postsynaptic depolarization: The duration of postsynaptic depolarization is determined by the binding of acetylcholine to its receptors and the subsequent removal of the neurotransmitter from the synapse. This is not directly affected by a defect in presynaptic calcium channels.

7. A 28-year-old man comes to the physician because of a 1-day history of blurred vision and difficulty swallowing. Physical examination shows that he has bilateral ptosis and generalized muscle weakness. He reports recently eating home-canned vegetables. It is suspected that a food-borne toxin is cleaving a protein involved in vesicular fusion. Which of the following transmission-related processes is most likely impaired in this patient?

- A. Presynaptic synthesis
- B. Exocytosis
- C. Reuptake of choline
- D. Postsynaptic binding
- E. Opening of voltage-gated calcium channels

Answer Key: B

Proteins of the SNARE complex are essential for the fusion of synaptic vesicles with the presynaptic membrane, a calcium-dependent process known as exocytosis. Cleavage of these proteins by proteases (like botulinum toxin) that may be ingested along with contaminated food can powerfully disrupt SNARE proteins, suppressing the release of transmitter at the neuromuscular junction and thereby causing weakness.

- A. Presynaptic synthesis: The toxin acts on proteins involved in vesicle fusion, not the synthesis of the neurotransmitter itself.
- C. Reuptake of choline: Reuptake mechanisms are not affected by a toxin that targets proteins associated with vesicular fusion.
- D. Postsynaptic binding: The toxin prevents neurotransmitter release, so there would be less acetylcholine to bind to the postsynaptic receptors. However, the binding process itself is not specifically inhibited.
- E. Opening of voltage-gated calcium channels: The toxin does not directly target the calcium channels that are responsible for calcium influx, but rather the proteins that act downstream of the calcium signal.

8. In an experimental setting involving cultured neurons, a ligand-gated receptor is mutated such that it functions as a metabotropic receptor (as opposed to an ionotropic receptor). Which of the following descriptions best applies to the altered pattern of neurotransmission in this experimental preparation?

- A. More rapid and transient
- B. G-protein-mediated
- C. Reuptake-independent
- D. Direct activation of ion channels
- E. Synaptic metabolism of transmitter

Answer Key: B

Metabotropic receptors are, by definition, G-protein-coupled receptors. Their activation initiates a cascade of intracellular signaling events that lead to a slow-onset and slow-offset postsynaptic response.

A. More rapid and transient: Ionotropic (not metabotropic) receptors mediate fast transient synaptic signaling.

C. Reuptake-independent: Both metabotropic and ionotropic synaptic contacts can be affected indirectly by uptake pumps.

D. Direct activation of ion channels: This is characteristic of the interaction of transmitter and ionotropic receptors. Metabotropic receptors, on the other hand, are not directly coupled to ion channels.

E. Synaptic metabolism of transmitter: Enzymatic degradation is a mechanism of neurotransmitter removal that is not dependent on the type of receptor present.

9. A 30-year-old woman comes to the physician because of the need for management of a mood disorder that is being treated with a serotonin-selective reuptake inhibitor (SSRI). Which of the following outcomes related to serotonergic transmission arises consequent to the administration of the medication in this patient?

A. Increased synthesis

B. Suppressed degradation

C. Increased synaptic longevity

D. Sensitization of receptors

E. Enhanced release

Answer Key: C

A serotonin-selective reuptake inhibitor inherently suppresses the removal of serotonin from the synaptic cleft, thereby increasing its synaptic longevity and its chances of interacting with synaptic receptors.

A. Increased synthesis: SSRIs do not directly affect synthesis; they affect the removal of the neurotransmitter from the synapse.

B. Suppressed degradation: Monoamine oxidase inhibitors (MAOIs) can suppress the enzymatic degradation of serotonin, but SSRIs do not.

D. Sensitization of receptors: While a prolonged presence of serotonin associated with the administration of SSRIs increases opportunities for

serotonin to bind to synaptic receptors, the SSRIs do not directly modulate the sensitivity of the receptors.

E. Enhanced release: SSRIs inhibit the removal of serotonin from synapses, but they do not directly affect mechanisms governing the release of the transmitter.

10. In an experimental setting involving cultured neurons, a defective mutation of the vesicle-associated transporter for dopamine is under study. Transport of which of the following ions is most likely impaired in these neurons?

- A. Sodium
- B. Potassium
- C. Chloride
- D. Calcium
- E. Hydrogen

Answer Key: E

Vesicle-associated transporters are responsible for actively pumping neurotransmitters from the cytoplasm into synaptic vesicles, a key step in their life cycle. The transporter exchanges cytoplasmic dopamine and intravesicular hydrogen. A defect in this transporter could therefore lead to impaired movement of hydrogen from the inside of vesicles to the cytoplasm.

A. Sodium: The relevant exchanger involves the outward movement of hydrogen but not sodium.

B. Potassium: The relevant exchanger involves the outward movement of hydrogen but not potassium.

C. Calcium: The relevant exchanger involves the outward movement of hydrogen but not calcium.

D. Chloride: The relevant exchanger involves the outward movement of hydrogen but not chloride.

11. A 22-year-old woman comes to the physician because of a 2-hour history of intense anxiety. The prescribed medication enhances the GABA-dependent influx of chloride, thereby reducing cellular excitability. Which of the following structures is most likely targeted by the drug?

- A. Metabotropic receptors
- B. Ionotropic receptors
- C. Voltage-gated potassium channels
- D. Voltage-gated sodium channels
- E. Acetylcholine receptors

Answer Key: B

Ionotropic receptors are ligand-gated ion channels that directly open in response to neurotransmitter binding. A drug that prolongs the opening of these channels (e.g., a benzodiazepine) would enhance their function. GABAA receptors bind GABA, leading to the near-immediate opening of a pore that permits the passage of chloride. The entry of chloride into neurons tends to repolarize or hyperpolarize neurons, with reduced neuronal excitability contributing to the anxiolytic effects of benzodiazepines.

- A. Metabotropic receptors: Metabotropic receptors mediate G-protein-dependent metabolic cascades that indirectly affect ion channels. They therefore do not directly control the duration of channel opening.
- C. Voltage-gated potassium channels: Voltage-gated potassium channels importantly limit cellular depolarization, but they are not directly linked to the function of chloride channels that are gated by GABA.
- D. Voltage-gated sodium channels: Voltage-gated sodium channels are associated with enhanced rather than diminished cellular excitability.
- E. Acetylcholine receptors: Acetylcholine receptors can be either ionotropic or metabotropic, but the medication interacts most intimately with GABAergic rather than cholinergic transmission.

12. A 6-month-old infant is brought to the physician because of a 1-day history of muscle weakness and poor feeding. The parents say that they recently changed to honey-sweetened pacifiers. Botulinum toxemia is suspected. Which of the following outcomes involving the neuromuscular junction is expected in this patient?
- A. Decreased quantal content
 - B. Diminished degradation of synaptic acetylcholine
 - C. Increased frequency of miniature end-plate potentials
 - D. Increased postsynaptic receptor density

E. Presynaptic hyperpolarization

Answer Key: A

Botulinum toxin inhibits neurotransmitter release, reducing the number of vesicles released per action potential (quantal content) at the neuromuscular junction. The endplate potential amplitude is proportional to quantal content. Endplate potentials are thus likely to be smaller in cases of botulinum toxemia.

B. Diminished degradation of synaptic acetylcholine: The toxin suppresses the release of acetylcholine. It does not directly impact enzymatic activity relevant to the synaptic longevity of acetylcholine.

C. Increased frequency of miniature end-plate potentials: The toxin also affects the spontaneous calcium-dependent release of acetylcholine, so the frequency of miniature endplate potentials would theoretically decrease rather than increase. Note, however, that miniature endplate potentials are not generally considered to be clinically relevant, as their robust demonstration is more typically restricted to experimental settings involving reduced extracellular calcium.

D. Increased postsynaptic receptor density: This could be a long-term adaptive response to denervation but not an immediate reaction to the toxin.

E. Presynaptic hyperpolarization: The toxin does not directly affect the resting membrane potential of the presynaptic terminal.

13. A 70-year-old man is brought to the physician because of a 6-month history of memory loss and cognitive decline. Alzheimer disease is suspected. An inhibitor of acetylcholinesterase is prescribed. Which of the following effects of the drug is most likely in this patient?
- A. Enhanced synthesis of acetylcholine
 - B. Inhibited reuptake of choline
 - C. Proliferation of postsynaptic acetylcholine receptors
 - D. Prolongation of the synaptic longevity of acetylcholine
 - E. Enhanced presynaptic calcium influx

Answer Key: D

Acetylcholinesterase inhibitors block the enzyme that breaks down acetylcholine, thereby increasing its concentration and prolonging its presence in the synapse.

- A. Enhanced synthesis of acetylcholine: The drug is an inhibitor of the enzyme that degrades acetylcholine, so it does not promote the synthesis of the transmitter.
- B. Inhibited reuptake of choline: The drug does not affect the reuptake pump. It targets the enzyme that degrades the neurotransmitter in the synapse.
- C. Proliferation of postsynaptic acetylcholine receptors: This could be a long-term adaptive response to denervation but not an immediate effect of the drug.
- E. Enhanced presynaptic calcium influx: This drug does not act on the presynaptic calcium channels found on presynaptic terminals. Rather, it acts on an enzyme in the synaptic cleft.

14. In an experimental setting involving cultured neurons, a genetic mutation causes dysfunction of a presynaptic protein involved in the docking of synaptic vesicles. This defect impairs the release of neurotransmitter. Which of the following outcomes is most likely in this experimental study?

- A. Increased amplitude of the postsynaptic potential
- B. Lowered threshold for the firing of action potentials
- C. Prevention of the postsynaptic potential
- D. Prolongation of the rising phase of the action potential
- E. Reduced amplitude of the postsynaptic potential

Answer Key: E

An impairment in docking and release would directly reduce the number of synaptic vesicles that fuse, thereby decreasing the amount of neurotransmitter released and reducing the amplitude of the postsynaptic potential.

- A. Increased amplitude of the postsynaptic potential: An impairment in docking would lead to a decrease in neurotransmitter release and thus a smaller postsynaptic potential.
- B. Lowered threshold for the firing action potentials: The mutation would exert its effects on the terminal end of the axon and not the electrical properties of the axon hillock.

C. Prevention of the postsynaptic potential: While this answer could be considered as correct, it is noteworthy that, while release is impaired, it need not be completely eliminated. Therefore, prevention of the postsynaptic potential could be considered as an extreme form of reduced amplitude of the postsynaptic potential.

D. Prolongation of the rising phase of the action potential: The dynamics of the rising phase of the action potential are determined chiefly by the electrical properties of the voltage-gated sodium channels. Deficiencies in release mechanisms related to transmitters ought not to impact the relevant population of voltage-gated channels.

15. A 16-year-old girl is brought to the physician because of a 6-month history of weakness. Laboratory studies show a genetic mutation affecting the enzyme that metabolizes the precursor molecule. Which of the following steps in the life cycle of the neurotransmitter is most likely affected in this patient?

- A. Reuptake
- B. Vesicular packaging
- C. Degradation
- D. Synthesis
- E. Retrograde transport

Answer Key: D

Small-molecule neurotransmitters, such as acetylcholine and dopamine, are synthesized in the presynaptic terminal through metabolism of a precursor. A genetic defect in a synthetic enzyme would directly impair this step, leading to a deficiency of the neurotransmitter.

A. Presynaptic reuptake: The defect is in synthesis, not reuptake.

B. Vesicular packaging: The neurotransmitter cannot be packaged into vesicles if it has not been synthesized.

C. Degradation: The enzyme for degradation works to remove the neurotransmitter, but the problem is that there is not enough of the neurotransmitter at the outset.

E. Retrograde transport: Retrograde transport involves the movement of materials from axonal terminals to the cell body. It is thus not a step in the synthesis of a small-molecule neurotransmitter.

